Topology Assignment Solutions

Topology: MAT307/M405/MAT803

Problem Sets 1–8

Submitted by **B23**

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Pula Karan	PS3: Q6–10
Dhrubojoti	PS4: Q1–4
Aaditya	PS4: Q5–8
Harisankar B	PS4: Q9–11
Prabhu	PS5: Q5–6
Yashpal Prasad	PS6: Q1–3
Indrajeet Kumawat	PS6: Q4–6
Himanshu Sharma	PS6: Q7–9
Karan Gangadharan	PS7: Q1–4

Coverage

Solved problems are shown in **green**; pending ones in grey.

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PS2	1 2 3 4 5 6 7 8 9 10 11 **12 13 14 15**	4/15
PS3	**1 2 3 4 5 6 7 8 9 10**	10/10
PS4	**1 2 3 4 5 6 7 8 9 10 11**	11/11
PS5	1 2 3 4 **5 6** 7 8 9 10	2/10
PS6	**1 2 3 4 5 6 7 8 9**	9/9
PS7	**1 2 3 4** 5 6	4/6
PS8	1 2 3 4 5 6 7 8 9 10	0/10
	Total solved	**50/88**

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Problem Set 1

Problem 1.1

Let A be a set and $\mathcal{P}(A)$ be its power set. Prove that there is no injective map $f\colon \mathcal{P}(A) \to A$ and no surjective map $g\colon A \to \mathcal{P}(A)$.

Solution *submitted by* **Ansh**

Proof. We establish both claims rigorously using Cantor's diagonalization argument, and we also show how the non-existence of an injection follows directly from the non-existence of a surjection.

Part I: Non-existence of a surjective map $g\colon A \to \mathcal{P}(A)$. Suppose, for the sake of contradiction, that there exists a surjective function $g\colon A \to \mathcal{P}(A)$. Define the diagonal subset $B \subseteq A$ by

$$B := \{x \in A \mid x \notin g(x)\}.$$

Note that this construction is well-defined for every set A (including $A = \emptyset$, in which case $\mathcal{P}(\emptyset) = \{\emptyset\} \neq \emptyset$, so no map $g\colon \emptyset \to \mathcal{P}(\emptyset)$ exists at all).

Since $B \subseteq A$, we have $B \in \mathcal{P}(A)$. Because g is assumed to be surjective, there exists at least one element $x_0 \in A$ such that

$$g(x_0) = B.$$

By the law of excluded middle, either $x_0 \in B$ or $x_0 \notin B$. We examine both cases:

(a) *Case 1 ($x_0 \in B$):* By the defining property of B, an element $x \in A$ belongs to B if and only if $x \notin g(x)$. Consequently, $x_0 \in B \implies x_0 \notin g(x_0)$. However, since $g(x_0) = B$, this implies $x_0 \notin B$, contradicting our assumption that $x_0 \in B$.

(b) *Case 2 ($x_0 \notin B$):* Since $g(x_0) = B$, the assumption $x_0 \notin B$ means $x_0 \notin g(x_0)$. By the definition of B, any element of A satisfying $x_0 \notin g(x_0)$ must belong to B; hence $x_0 \in B$, contradicting our assumption that $x_0 \notin B$.

In both cases we arrive at the logical contradiction $x_0 \in B \iff x_0 \notin B$. Therefore, our initial assumption is false, and no surjective map $g\colon A \to \mathcal{P}(A)$ exists.

Part II: Non-existence of an injective map $f\colon \mathcal{P}(A) \to A$. We provide two independent proofs of this result.

Method 1 (Direct Diagonal Proof). Suppose, for the sake of contradiction, that there exists an injective map $f\colon \mathcal{P}(A) \to A$. Consider the subset $C \subseteq A$ defined by

$$C := \{f(S) \in A \mid S \in \mathcal{P}(A) \text{ and } f(S) \notin S\}.$$

Since $C \subseteq A$, we have $C \in \mathcal{P}(A)$. Let $c := f(C) \in A$. We ask whether $c \in C$ or $c \notin C$:

(a) *Case 1 ($c \in C$):* By definition of C, there exists some subset $S \in \mathcal{P}(A)$ such that $c = f(S)$ and $f(S) \notin S$. Since $f(C) = c = f(S)$ and f is injective, it follows that $C = S$. Substituting $S = C$ into $f(S) \notin S$ yields $c = f(C) \notin C$, a contradiction.

(b) *Case 2 ($c \notin C$):* Since $c = f(C)$, the statement $c \notin C$ is equivalent to $f(C) \notin C$. Taking $S = C \in \mathcal{P}(A)$ in the defining condition for C, we see that $f(C)$ satisfies the membership criterion for C, which implies $c = f(C) \in C$, a contradiction.

Thus, in both cases we obtain a contradiction. Hence, no injective map $f\colon \mathcal{P}(A) \to A$ can exist.

Method 2 (Reduction to Part I). Suppose $f\colon \mathcal{P}(A) \to A$ is injective. Since $\mathcal{P}(A)$ is always non-empty (as $\emptyset \in \mathcal{P}(A)$), its image $f(\mathcal{P}(A)) \subseteq A$ is non-empty, which implies $A \neq \emptyset$. Define a map $g\colon A \to \mathcal{P}(A)$ by

$$g(a) := \begin{cases} S & \text{if } a = f(S) \text{ for some } S \in \mathcal{P}(A), \\ \emptyset & \text{if } a \notin f(\mathcal{P}(A)). \end{cases}$$

Because f is injective, for each $a \in f(\mathcal{P}(A))$ the preimage $S \in \mathcal{P}(A)$ with $f(S) = a$ is unique; thus g is a well-defined function. Moreover, for every $S \in \mathcal{P}(A)$, we have $g(f(S)) = S$, showing that $g\colon A \to \mathcal{P}(A)$ is surjective. This directly contradicts Part I. Therefore, no injective map $f\colon \mathcal{P}(A) \to A$ exists. □

Problem 1.2

Let X be an infinite set. Is the collection $\mathcal{T}_\infty := \{U \subseteq X \mid X \setminus U \text{ is infinite or } U = \emptyset\}$ a topology on X? Justify your answer.

Solution

submitted by **Ansh**

Answer. No, τ_∞ is not a topology on X. It violates both the first axiom ($X \in \tau_\infty$) and the second axiom (closure under arbitrary unions; in fact, it even fails closure under finite unions).

Justification. For τ_∞ to be a topology on X, it must satisfy the three standard topological axioms:

(T1) $\emptyset \in \tau_\infty$ and $X \in \tau_\infty$;

(T2) τ_∞ is closed under arbitrary unions;

(T3) τ_∞ is closed under finite intersections.

We examine each axiom in turn:

1. Violation of Axiom (T1) ($X \notin \tau_\infty$). By explicit definition, $\emptyset \in \tau_\infty$. However, consider the whole space $U = X$. Since X is an infinite set, $X \neq \emptyset$. Furthermore, the complement of X in X is

$$X \setminus X = \emptyset,$$

which has cardinality 0 and is therefore a finite set, not an infinite set. Because $U = X$ neither equals $\emptyset$ nor has an infinite complement in X, we have

$$X \notin \tau_\infty.$$

Thus, Axiom (T1) is violated.

2. Violation of Axiom (T2) (Failure of closure under unions). Even if one were to adjoin X to the collection, τ_∞ fails to be closed under arbitrary unions—and even fails under the union of two sets:

- **Counterexample via arbitrary unions of singletons:** For any element $x \in X$, consider the singleton $U_x = \{x\}$. Since X is infinite, removing a single element leaves an infinite set, so $X \setminus \{x\}$ is infinite and $\{x\} \in \tau_\infty$ for every $x \in X$. Fix a point $x_0 \in X$ and consider the union

$$U := \bigcup_{x \in X \setminus \{x_0\}} \{x\} = X \setminus \{x_0\}.$$

Since X is infinite, $U = X \setminus \{x_0\}$ is non-empty (and $U \neq X$). Yet its complement is $X \setminus U = \{x_0\}$, which is a finite set of cardinality 1. Therefore, $U \notin \tau_\infty$.

- **Counterexample via a finite union of two sets:** Since X is infinite, X contains a countably infinite subset $C = \{x_1, x_2, x_3, \dots\}$ with $x_i \neq x_j$ for $i \neq j$. Partition C into the set of even-indexed elements $E = \{x_2, x_4, x_6, \dots\}$ and odd-indexed elements $O = \{x_1, x_3, x_5, \dots\}$, both of which are infinite. Define

$$U_1 := X \setminus (E \cup \{x_1\}) \quad \text{and} \quad U_2 := X \setminus O.$$

Then $X \setminus U_1 = E \cup \{x_1\}$ is infinite, so $U_1 \in \tau_\infty$; similarly, $X \setminus U_2 = O$ is infinite, so $U_2 \in \tau_\infty$. However, by De Morgan's laws,

$$U_1 \cup U_2 = X \setminus \big((E \cup \{x_1\}) \cap O\big) = X \setminus \{x_1\}.$$

Since $U_1 \cup U_2 = X \setminus \{x_1\} \neq \emptyset$ and its complement $X \setminus (U_1 \cup U_2) = \{x_1\}$ is finite (of size 1), we have $U_1 \cup U_2 \notin \tau_\infty$. (Moreover, taking $V_1 = X \setminus E \in \tau_\infty$ and $V_2 = X \setminus O \in \tau_\infty$ gives $V_1 \cup V_2 = X \notin \tau_\infty$.)

3. Remark on Axiom (T3) (Closure under finite intersections holds). For completeness, note that if $U_1, U_2 \in \tau_\infty$, then either one of them is $\emptyset$ (so $U_1 \cap U_2 = \emptyset \in \tau_\infty$) or both $X \setminus U_1$ and $X \setminus U_2$ are infinite, in which case $X \setminus (U_1 \cap U_2) = (X \setminus U_1) \cup (X \setminus U_2)$ contains $X \setminus U_1$ and is therefore infinite, so $U_1 \cap U_2 \in \tau_\infty$. Nevertheless, because Axioms (T1) and (T2) fail, τ_∞ is not a topology on X.

Problem 1.3

Let $X = \{a, b, c\}$. Determine which of the following collections of subsets form a valid topology on X. For those that fail, explicitly state which axiom or axioms are violated:

(i) $\mathcal{T}_1 = \{\emptyset, \{a\}, \{b\}, X\}$;

(ii) $\mathcal{T}_2 = \{\emptyset, \{a\}, \{a, b\}, X\}$;

(iii) $\mathcal{T}_3 = \{\emptyset, \{a, b\}, \{b, c\}, X\}$.

Solution

submitted by **Ansh**

Recall that a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ is a topology on X if and only if: (T1) $\emptyset, X \in \mathcal{T}$; (T2) $\mathcal{T}$ is closed under arbitrary unions; and (T3) $\mathcal{T}$ is closed under finite intersections.

Part (i): $\mathcal{T}_1 = \{\emptyset, \{a\}, \{b\}, X\}$.

- **Axiom (T1):** Satisfied, since $\emptyset \in \mathcal{T}_1$ and $X \in \mathcal{T}_1$.
- **Axiom (T2) (Arbitrary Unions): Violated.** Taking the union of $\{a\} \in \mathcal{T}_1$ and $\{b\} \in \mathcal{T}_1$ yields

$$\{a\} \cup \{b\} = \{a, b\} \notin \mathcal{T}_1.$$

- **Axiom (T3) (Finite Intersections):** Satisfied, since $\{a\} \cap \{b\} = \emptyset \in \mathcal{T}_1$, and the intersection of any element of $\mathcal{T}_1$ with $\emptyset$, X, or itself remains in $\mathcal{T}_1$.
- **Conclusion:** $\mathcal{T}_1$ is **not a valid topology** on X because it violates the **closure under unions axiom (T2)**.

Part (ii): $\mathcal{T}_2 = \{\emptyset, \{a\}, \{a,b\}, X\}$. Observe that the sets in $\mathcal{T}_2$ form a strictly nested chain under set inclusion:

$$\emptyset \subsetneq \{a\} \subsetneq \{a,b\} \subsetneq \{a,b,c\} = X.$$

- **Axiom (T1):** Satisfied, since $\emptyset \in \mathcal{T}_2$ and $X \in \mathcal{T}_2$.
- **Axiom (T2) (Arbitrary Unions):** Satisfied. Because $\mathcal{T}_2$ is finite and totally ordered by inclusion ($U \subseteq V$ or $V \subseteq U$ for all $U, V \in \mathcal{T}_2$), the union of any non-empty subcollection of $\mathcal{T}_2$ equals the largest element of that subcollection, which belongs to $\mathcal{T}_2$ (and the empty union is $\emptyset \in \mathcal{T}_2$). Explicitly, the only non-trivial pairwise union is $\{a\} \cup \{a,b\} = \{a,b\} \in \mathcal{T}_2$.
- **Axiom (T3) (Finite Intersections):** Satisfied. For any non-empty finite subcollection of $\mathcal{T}_2$, its intersection equals the smallest element in that subcollection, which belongs to $\mathcal{T}_2$. Explicitly, $\{a\} \cap \{a,b\} = \{a\} \in \mathcal{T}_2$.
- **Conclusion: $\mathcal{T}_2$ is a valid topology** on X.

Part (iii): $\mathcal{T}_3 = \{\emptyset, \{a,b\}, \{b,c\}, X\}$.

- **Axiom (T1):** Satisfied, since $\emptyset \in \mathcal{T}_3$ and $X \in \mathcal{T}_3$.
- **Axiom (T2) (Arbitrary Unions):** Satisfied, because $\{a,b\} \cup \{b,c\} = \{a,b,c\} = X \in \mathcal{T}_3$, and any union involving $\emptyset$, X, or a single set remains in $\mathcal{T}_3$.
- **Axiom (T3) (Finite Intersections): Violated.** Taking the intersection of $\{a,b\} \in \mathcal{T}_3$ and $\{b,c\} \in \mathcal{T}_3$ yields

$$\{a,b\} \cap \{b,c\} = \{b\} \notin \mathcal{T}_3.$$

- **Conclusion:** $\mathcal{T}_3$ is **not a valid topology** on X because it violates the **closure under finite intersections axiom (T3)**.

Collection	(T1)	(T2) Unions	(T3) Intersections	Topology?
(i) $\mathcal{T}_1$	Holds	Fails ($\{a\} \cup \{b\} = \{a,b\} \notin \mathcal{T}_1$)	Holds	**No**
(ii) $\mathcal{T}_2$	Holds	Holds	Holds	**Yes**
(iii) $\mathcal{T}_3$	Holds	Holds	Fails ($\{a,b\} \cap \{b,c\} = \{b\} \notin \mathcal{T}_3$)	**No**

Problem 1.4

Let X be a topological space and $A \subset X$. Prove that A is open in X if and only if for each $x \in A$, $\exists$ an open set U in X such that $x \in U \subseteq A$.

Solution

submitted by **Ansh**

Proof. Let (X, τ) be a topological space and let $A \subseteq X$. We prove both directions of the biconditional statement.

Forward Direction ($\Longrightarrow$). Suppose that A is open in X, so $A \in \tau$. Let $x \in A$ be arbitrary. Choose $U := A$. Since A is open in X, U is an open set in X, and by reflexivity of set inclusion ($A \subseteq A$), we immediately have

$$x \in U \subseteq A.$$

Thus, for each $x \in A$, there exists an open set U in X such that $x \in U \subseteq A$.

Reverse Direction ($\Longleftarrow$). Conversely, suppose that for every $x \in A$, there exists an open set $U_x \in \tau$ such that $x \in U_x \subseteq A$. We claim that

$$A = \bigcup_{x \in A} U_x.$$

We establish this set equality by proving mutual inclusion:

1. *Proof that* $A \subseteq \bigcup_{x \in A} U_x$*:* Let $y \in A$ be arbitrary. By hypothesis, the corresponding open set U_y satisfies $y \in U_y$. Since U_y is one of the sets in the indexed family $\{U_x\}_{x \in A}$, it follows from the definition of the union that $y \in \bigcup_{x \in A} U_x$. Hence, $A \subseteq \bigcup_{x \in A} U_x$. (Note that if $A = \emptyset$, the empty union is $\emptyset$, so the inclusion holds vacuously.)
2. *Proof that* $\bigcup_{x \in A} U_x \subseteq A$*:* Let $y \in \bigcup_{x \in A} U_x$ be arbitrary. By definition of the union, there exists some index $x_0 \in A$ such that $y \in U_{x_0}$. By hypothesis, $U_{x_0} \subseteq A$, which implies $y \in A$. Hence, $\bigcup_{x \in A} U_x \subseteq A$.

Combining both inclusions yields $A = \bigcup_{x \in A} U_x$. Because each U_x is an open set in X and a topology τ is closed under arbitrary unions, we conclude that $A = \bigcup_{x \in A} U_x \in \tau$; that is, A is open in X. □

Problem 1.5

Let U be an open set and V be a closed set in X. Prove that $U \setminus V$ is open in X and $V \setminus U$ is closed in X.

Solution

submitted by **Ansh**

Proof. Let (X, τ) be a topological space, let $U \subseteq X$ be an open set in X, and let $V \subseteq X$ be a closed set in X.

First, recall that for any two subsets $A, B \subseteq X$, the set difference $A \setminus B$ is given by the identity

$$A \setminus B = \{x \in X \mid x \in A \text{ and } x \notin B\} = A \cap (X \setminus B).$$

Furthermore, by the definition of a closed set in a topological space, a subset $F \subseteq X$ is closed in X if and only if its complement $X \setminus F$ is open in X.

Part I: Proof that $U \setminus V$ is open in X. Applying the set-difference identity to $U \setminus V$, we have

$$U \setminus V = U \cap (X \setminus V).$$

Since V is closed in X, its complement $X \setminus V$ is open in X by definition. By hypothesis, U is also open in X. By the finite intersection axiom of a topology, the intersection of two open sets in X is open in X. Therefore, $U \setminus V = U \cap (X \setminus V)$ is open in X.

Part II: Proof that $V \setminus U$ is closed in X. We provide both a direct proof from the open-set axioms via De Morgan's laws and a dual proof via closed-set intersections:

Method 1 (Direct proof via De Morgan's Laws). To show that $V \setminus U$ is closed in X, it suffices by definition to prove that its complement $X \setminus (V \setminus U)$ is open in X. Using $V \setminus U = V \cap (X \setminus U)$

and De Morgan's laws, we compute:

$$\begin{aligned} X \setminus (V \setminus U) &= X \setminus \big(V \cap (X \setminus U)\big) \\ &= (X \setminus V) \cup \big(X \setminus (X \setminus U)\big) \\ &= (X \setminus V) \cup U. \end{aligned}$$

Since V is closed in X, its complement $X \setminus V$ is open in X. By hypothesis, U is also open in X. Because a topology is closed under unions, $(X \setminus V) \cup U$ is open in X. Consequently, $X \setminus (V \setminus U)$ is open in X, which by definition means that $V \setminus U$ is closed in X.

Method 2 (Via intersections of closed sets). Using the set-difference identity, we write

$$V \setminus U = V \cap (X \setminus U).$$

Since U is open in X and $X \setminus (X \setminus U) = U$, the set $X \setminus U$ is closed in X. We are also given that V is closed in X. Since the intersection of any two closed sets in X is closed in X, it follows immediately that $V \setminus U = V \cap (X \setminus U)$ is closed in X.

This completes the proof of both statements. □

Problem 1.6

Let $X = \mathbb{N}$ and define $\mathcal{T}_1, \mathcal{T}_2 \subseteq \mathcal{P}(\mathbb{N})$ as follows:

(i) $\mathcal{T}_1 := \{\emptyset, \mathbb{N}, \{1\}, \{1,2\}, \{1,2,3\}, \{1,2,3,4\}, \dots\}$.

(ii) $\mathcal{T}_2 := \{\emptyset, \mathbb{N}, \mathbb{N} \setminus \{1\}, \mathbb{N} \setminus \{1,2\}, \mathbb{N} \setminus \{1,2,3\}, \mathbb{N} \setminus \{1,2,3,4\}, \dots\}$.

Determine whether $\mathcal{T}_1$ and $\mathcal{T}_2$ define topologies on $\mathbb{N}$.

Solution

submitted by **Sahil**

Throughout, a topology on a set X is a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ such that (1) $\varnothing, X \in \mathcal{T}$, (2) arbitrary unions of members of $\mathcal{T}$ belong to $\mathcal{T}$, and (3) finite intersections of members of $\mathcal{T}$ belong to $\mathcal{T}$.

Let

$$A_n = \{1, 2, \dots, n\}, \qquad B_n = \mathbb{N} \setminus A_n \quad (n \geq 1).$$

Then the two collections in the question can be written as

$$\mathcal{T}_1 = \{\varnothing, \mathbb{N}\} \cup \{A_n : n \geq 1\}, \qquad \mathcal{T}_2 = \{\varnothing, \mathbb{N}\} \cup \{B_n : n \geq 1\}.$$

We verify the topology axioms separately.

(a) The collection $\mathcal{T}_1$ is a topology. Clearly, $\varnothing, \mathbb{N} \in \mathcal{T}_1$. For finite intersections, observe that

$$A_m \cap A_n = A_{\min\{m,n\}}.$$

Intersections with $\varnothing$ or $\mathbb{N}$ also belong to $\mathcal{T}_1$. Thus $\mathcal{T}_1$ is closed under finite intersections.

Now let $\{U_i\}_{i \in I}$ be any family of members of $\mathcal{T}_1$. If every U_i is empty, then $\bigcup_{i \in I} U_i = \varnothing$. If at least one $U_i = \mathbb{N}$, the union is $\mathbb{N}$. In the remaining case, the nonempty sets in the family are of the form A_n. If their indices are unbounded in $\mathbb{N}$, then every positive integer belongs to some A_n, so their union is $\mathbb{N}$. If their indices are bounded, they have a largest value m (a nonempty bounded subset of $\mathbb{N}$ has a maximum), and the union is A_m. In every case the union belongs to $\mathcal{T}_1$. Hence $\mathcal{T}_1$ is a topology on $\mathbb{N}$.

(b) The collection $\mathcal{T}_2$ is a topology. Clearly, $\varnothing, \mathbb{N} \in \mathcal{T}_2$. For finite intersections, De Morgan's law gives

$$B_m \cap B_n = (\mathbb{N} \setminus A_m) \cap (\mathbb{N} \setminus A_n) = \mathbb{N} \setminus (A_m \cup A_n) = B_{\max\{m,n\}}.$$

Intersections involving $\varnothing$ or $\mathbb{N}$ also remain in $\mathcal{T}_2$.

Consider an arbitrary family $\{U_i\}_{i \in I} \subseteq \mathcal{T}_2$. If one member is $\mathbb{N}$, its union is $\mathbb{N}$. If every member is empty, its union is empty. Otherwise, discard the empty members and write the remaining members as B_{n_i}. The set of indices $\{n_i : i \in I,\ U_i \neq \varnothing\}$ is a nonempty subset of $\mathbb{N}$, so it has a least element, say m. Since the sets B_n increase as n decreases, we have

$$\bigcup_{i \in I} B_{n_i} = B_m.$$

Indeed, B_m is one of the sets in the union and contains every B_{n_i} because $m \leq n_i$. Thus the union belongs to $\mathcal{T}_2$. Therefore $\mathcal{T}_2$ is also a topology on $\mathbb{N}$.

Conclusion: Both $\mathcal{T}_1$ and $\mathcal{T}_2$ define topologies on $\mathbb{N}$.

Problem 1.7

Show that if $\mathscr{B}$ is a basis for a topology on X, then the topology generated by $\mathscr{B}$ equals the intersection of all topologies on X that contain $\mathscr{B}$.

Solution

submitted by **Sahil**

Let $\mathcal{B}$ be a basis for a topology on X, and let $\mathcal{T}_{\mathcal{B}}$ denote the topology generated by $\mathcal{B}$. By the definition of the topology generated by a basis,

$$\mathcal{T}_{\mathcal{B}} = \Big\{ \bigcup_{\alpha \in A} B_\alpha : A \text{ is an index set and } B_\alpha \in \mathcal{B} \Big\}.$$

Let $\mathfrak{F}$ be the collection of all topologies on X that contain $\mathcal{B}$. We prove

$$\boxed{\mathcal{T}_{\mathcal{B}} = \bigcap_{\mathcal{T} \in \mathfrak{F}} \mathcal{T}.}$$

First, let $U \in \mathcal{T}_{\mathcal{B}}$. Then U is a union of elements of $\mathcal{B}$. If $\mathcal{T} \in \mathfrak{F}$, then $\mathcal{B} \subseteq \mathcal{T}$. Since every topology is closed under arbitrary unions, it follows that $U \in \mathcal{T}$. This holds for every $\mathcal{T} \in \mathfrak{F}$, and hence

$$\mathcal{T}_{\mathcal{B}} \subseteq \bigcap_{\mathcal{T} \in \mathfrak{F}} \mathcal{T}.$$

For the reverse inclusion, note that $\mathcal{T}_{\mathcal{B}}$ is itself a topology on X containing $\mathcal{B}$. Consequently, $\mathcal{T}_{\mathcal{B}} \in \mathfrak{F}$. The intersection of all members of $\mathfrak{F}$ is therefore contained in this particular member:

$$\bigcap_{\mathcal{T} \in \mathfrak{F}} \mathcal{T} \subseteq \mathcal{T}_{\mathcal{B}}.$$

Combining the two inclusions proves the required equality.

Interpretation: The topology generated by a basis is the smallest topology on X that contains every basis element.

Problem 1.8

Let X be an infinite set and $\mathcal{T}_f$ be the finite complement topology on X. If U and V are non-empty open sets in $(X, \mathcal{T}_f)$, prove that $U \cap V \neq \emptyset$.

Solution *submitted by* **Sahil**

Let X be infinite and let

$$\mathcal{T}_f = \{\varnothing\} \cup \{U \subseteq X : X \setminus U \text{ is finite}\}$$

be the finite-complement topology. Suppose $U, V \in \mathcal{T}_f$ are nonempty. By the definition of $\mathcal{T}_f$, both complements $X \setminus U$ and $X \setminus V$ are finite. Using De Morgan's law,

$$X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V).$$

The right-hand side is a union of two finite sets, and is therefore finite. If $U \cap V = \varnothing$, then the left-hand side equals X, which would imply that X is finite. This contradicts the assumption that X is infinite. Hence

$$\boxed{U \cap V \neq \varnothing.}$$

Problem 1.9

Let $X \neq \emptyset$ be a set and let $a \in X$ be a fixed point. Prove that the *particular point* collection

$$\mathcal{T}_a = \{V \subseteq X \mid V = \emptyset \text{ or } a \in V\}$$

defines a valid topology on X.

Solution *submitted by* **Sahil**

Let $X \neq \varnothing$ and fix $a \in X$. Define

$$\mathcal{T}_a = \{V \subseteq X : V = \varnothing \text{ or } a \in V\}.$$

We check the topology axioms.

(i) Empty set and whole space. By definition, $\varnothing \in \mathcal{T}_a$. Also $a \in X$, so $X \in \mathcal{T}_a$.

(ii) Arbitrary unions. Let $\{V_i\}_{i \in I}$ be any family in $\mathcal{T}_a$. If every V_i is empty, then $\bigcup_{i \in I} V_i = \varnothing$. Otherwise, at least one member, say V_j, is nonempty. Since $V_j \in \mathcal{T}_a$, it contains a. Therefore $a \in \bigcup_{i \in I} V_i$, so the union belongs to $\mathcal{T}_a$.

(iii) Finite intersections. It is enough to check intersections of two members, since repeated application gives the result for every finite intersection. Let $U, V \in \mathcal{T}_a$. If either set is empty, then $U \cap V = \varnothing \in \mathcal{T}_a$. If neither is empty, both contain a, and hence $a \in U \cap V$. Thus $U \cap V \in \mathcal{T}_a$.

All three axioms hold. Therefore $\boxed{\mathcal{T}_a \text{ is a topology on } X.}$

Problem 1.10

Let $X \neq \emptyset$ be a set and let $a \in X$ be a fixed point. Prove that the *excluded point* collection

$$\mathcal{T}^a := \{V \subseteq X \mid V = X \text{ or } a \notin V\}$$

defines a valid topology on X.

Solution

submitted by **Sahil**

Let $X \neq \varnothing$ and fix $a \in X$. Define

$$\mathcal{T}_a^e = \{V \subseteq X : V = X \text{ or } a \notin V\}.$$

Thus the open sets are X itself and all subsets of X that do not contain a. We verify the topology axioms.

(i) Empty set and whole space. Since $a \notin \varnothing$, we have $\varnothing \in \mathcal{T}_a^e$. Also $X \in \mathcal{T}_a^e$ by definition.

(ii) Arbitrary unions. Let $\{V_i\}_{i \in I}$ be any family in $\mathcal{T}_a^e$. If some $V_j = X$, then $\bigcup_{i \in I} V_i = X$. If no member equals X, then every V_i omits a. Consequently, their union also omits a:

$$a \notin V_i \text{ for all } i \in I \quad \Longrightarrow \quad a \notin \bigcup_{i \in I} V_i.$$

In either case the union belongs to $\mathcal{T}_a^e$.

(iii) Finite intersections. Again it is enough to consider two sets. Let $U, V \in \mathcal{T}_a^e$. If both equal X, then $U \cap V = X$. Otherwise, at least one of them, say U, is not X. By the definition of the collection, $a \notin U$. Since $U \cap V \subseteq U$, the intersection also omits a. Hence $U \cap V \in \mathcal{T}_a^e$.

All three topology axioms are satisfied. Therefore

$$\boxed{\mathcal{T}_a^e \text{ is a topology on } X.}$$

Problem 1.11

Let X be an infinite set and let $a \in X$ be a fixed point. Define a collection of subsets $\mathcal{T}_a \subseteq \mathcal{P}(X)$ by:

$$\mathcal{T}_a = \mathcal{P}(X \setminus \{a\}) \cup \{V \subseteq X \mid a \in V \text{ and } X \setminus V \text{ is finite}\}.$$

Prove that $\mathcal{T}_a$ defines a valid topology on X. What happens when X is a finite set?

Solution pending

Problem 1.12

Let X be an infinite set and let $a, b \in X$ be two distinct points. Define a collection of subsets $\mathcal{T} \subseteq \mathcal{P}(X)$ by:

$$\mathcal{T} = \mathcal{P}(X \setminus \{a, b\}) \cup \{V \subseteq X \mid a \in V \text{ or } b \in V \text{ and } X \setminus V \text{ is finite}\}.$$

Is $\mathcal{T}$ a topology on X?

Solution pending

Problem 1.13

Let $\mathscr{B}_1$, $\mathscr{B}_2$, and $\mathscr{B}_3$ be respectively the collections of all open balls, open rectangles, and open triangles in the plane $\mathbb{R}^2$.

(i). Prove that all three collections satisfy the basis axioms on $\mathbb{R}^2$.

(ii). Let $\mathcal{T}_i$ be the topology generated by the basis $\mathscr{B}_i$ for $i = 1, 2, 3$. Let $f\colon (\mathbb{R}^2, \mathcal{T}_i) \to (\mathbb{R}^2, \mathcal{T}_j)$ be the identity map. Prove that for any choice of $i, j \in \{1, 2, 3\}$, the map f is continuous.

(iii). Conclude that $\mathcal{T}_1 = \mathcal{T}_2 = \mathcal{T}_3$.

Solution pending

Problem 1.14

Along with the finite complement topology ($\mathcal{T}_f$) and the countable complement topology ($\mathcal{T}_c$), we consider the topologies generated by the following six bases:

(i) **Standard Topology ($\mathcal{T}_{\text{std}}$):** Generated by the basis of all open intervals:

$$\mathscr{B}_1 = \{(a, b) \mid a < b \text{ and } a, b \in \mathbb{R}\}.$$

(ii) **K-Topology ($\mathcal{T}_K$):** Generated by the basis:

$$\mathscr{B}_2 = \{(a, b) \mid a < b\} \cup \{(a, b) \setminus K \mid a < b\}, \quad \text{where } K = \{1, \tfrac{1}{2}, \tfrac{1}{3}, \dots\}.$$

(iii) **Lower Limit Topology ($\mathcal{T}_\ell$):** Generated by the basis of all half-open intervals closed on the left:

$$\mathscr{B}_3 = \{[a, b) \mid a < b \text{ and } a, b \in \mathbb{R}\}.$$

(iv) **Upper Limit Topology ($\mathcal{T}_u$):** Generated by the basis of all half-open intervals closed on the right:

$$\mathscr{B}_4 = \{(a, b] \mid a < b \text{ and } a, b \in \mathbb{R}\}.$$

(v) **Left-Ray Topology ($\mathcal{T}_{\text{left}}$):** Generated by the basis of open left rays:

$$\mathscr{B}_5 = \{(-\infty, b) \mid b \in \mathbb{R}\}.$$

(vi) **Right-Ray Topology ($\mathcal{T}_{\text{right}}$):** Generated by the basis of open right rays:

$$\mathscr{B}_6 = \{(a, \infty) \mid a \in \mathbb{R}\}.$$

For each pair of these topologies, determine whether one contains the other (i.e., whether one is finer, coarser, or if they are completely incomparable). Summarize your results by drawing a partial order hierarchy diagram representing the containment relations among all eight topologies.

Solution pending

Problem 1.15

Let $\mathcal{T}_X$ denote the collection of all topologies on a non-empty set X and let $\mathcal{T}_X/\text{hom}$ be the set of homeomorphism classes of topologies on X. Let S_X be the group of all bijective maps on X. Note that each bijection $f\colon X \to X$ induces a bijection $\tilde{f}\colon \mathcal{P}(X) \to \mathcal{P}(X)$ defined by $\tilde{f}(A) = \{f(a) \mid a \in A\}$ for every subset $A \in \mathcal{P}(X)$. Consequently, S_X acts naturally on $\mathcal{T}_X$ via $f.\mathcal{T} = \tilde{f}(\mathcal{T})$ for all $f \in S_X$ and $\mathcal{T} \in \mathcal{T}_X$. Prove that there is a natural bijection between $\mathcal{T}_X/\text{hom}$ and the collection of orbits of the natural action of S_X on $\mathcal{T}_X$.

Solution pending

Problem 1.16

List down all possible (there are 29) topologies on a three-point set $X = \{a, b, c\}$. Compute the number of orbits under the natural action of S_3 on the set $\mathcal{T}_X$ of all topologies on X.

Solution pending

Problem 1.17

Let X be a set, and let $\mathcal{S}_1$ and $\mathcal{S}_2$ be two subbases on X generating topologies $\mathcal{T}_1$ and $\mathcal{T}_2$, respectively.

(i). If $\mathcal{S}_1 \subseteq \mathcal{S}_2$, prove that $\mathcal{T}_1 \subseteq \mathcal{T}_2$.

(ii). Give an explicit example where $\mathcal{T}_1 = \mathcal{T}_2$ even though $\mathcal{S}_1 \neq \mathcal{S}_2$.

Solution pending

Problem Set 2

Problem 2.1

Whether each of the following subsets of $\mathbb{R}^2$ can be written as a Cartesian product $A \times B$, where $A, B \subset \mathbb{R}$?

(i) $\{(x, y) \mid x \text{ is an integer}\}$.

(ii) $\{(x, y) \mid 0 \leq x < 1\}$.

(iii) $\{(x, y) \mid x < y\}$.

(iv) $\{(x, y) \mid x \text{ is an integer, and } y \text{ is not a rational number}\}$.

(v) $\{(x, y) \mid x^2 + y^2 \leq 2026\}$.

Solution pending

Problem 2.2

Let $f \colon A \to B$ be a function, with $A_0 \subseteq A$ and $B_0 \subseteq B$.

(i) Show that $A_0 \subseteq f^{-1}(f(A_0))$, and that equality holds for every subset $A_0 \subseteq A$ if and only if f is injective.

(ii) Show that $f(f^{-1}(B_0)) \subseteq B_0$, and that equality holds for every subset $B_0 \subseteq B$ if and only if f is surjective.

Solution pending

Problem 2.3

Let $f \colon A \to B$ be a function, and $B_0, B_1 \subseteq B$. Show that preimages preserve inclusions, unions, intersections, and set differences by proving the following:

(i) If $B_0 \subseteq B_1$, then $f^{-1}(B_0) \subseteq f^{-1}(B_1)$.

(ii) $f^{-1}(B_0 \cup B_1) = f^{-1}(B_0) \cup f^{-1}(B_1)$.

(iii) $f^{-1}(B_0 \cap B_1) = f^{-1}(B_0) \cap f^{-1}(B_1)$.

(iv) $f^{-1}(B_0 \setminus B_1) = f^{-1}(B_0) \setminus f^{-1}(B_1)$.

Solution pending

Problem 2.4

Let $f\colon A \to B$ be a function, and let $A_0, A_1 \subseteq A$.

(i) Show that images preserve inclusions and unions:

(a) If $A_0 \subseteq A_1$, then $f(A_0) \subseteq f(A_1)$.

(b) $f(A_0 \cup A_1) = f(A_0) \cup f(A_1)$.

(ii) Show that images satisfy only one-way inclusions for intersections and set differences in general:

(a) $f(A_0 \cap A_1) \subseteq f(A_0) \cap f(A_1)$.

(b) $f(A_0) \setminus f(A_1) \subseteq f(A_0 \setminus A_1)$.

(iii) Prove that equality holds in both (ii)(a) and (ii)(b) for all subsets $A_0, A_1 \subseteq A$ if and only if f is injective.

Solution pending

Problem 2.5

Let $\sim$ be a relation defined on the Euclidean plane $\mathbb{R}^2$ as follows: two points are related if and only if their distances from the origin are equal.

(i) Show that $\sim$ is an equivalence relation (i.e., reflexivity, symmetry, and transitivity).

(ii) Describe the equivalence class $[P]$ for any point $P \in \mathbb{R}^2$.

(iii) What is the geometric interpretation of the partition of $\mathbb{R}^2$ induced by this relation?

Solution pending

Problem 2.6

Let X be a non-empty set. Suppose that S is the collection of the following two types of open rays: $(a, \infty) = \{x \in X \mid a < x\}$, and $(-\infty, b) = \{x \in X \mid x < b\}$. Prove that S is a subbasis, and it generates the order topology on X.

Solution pending

Problem 2.7

Let $\{A_n\}_{n=1}^{\infty}$ be a sequence of non-empty countable sets.

(1) Prove that the finite Cartesian product $\prod_{i=1}^{k} A_i = A_1 \times A_2 \times \cdots \times A_k$ is countable for any $k \in \mathbb{N}$.

(2) Determine whether the infinite Cartesian product $\prod_{n=1}^{\infty} A_n$ is necessarily countable. Justify your answer.

Solution pending

Problem 2.8

Let X be a linearly ordered set. Prove that a singleton set $\{x\}$ is open in the order topology on X if and only if x has an immediate predecessor and an immediate successor, or if x is the minimum/maximum and has an immediate successor/predecessor respectively.

Solution pending

Problem 2.9

Prove that if V is closed in Y and Y is closed in X, then V is closed in X.

Solution pending

Problem 2.10

Let Y be a subspace of X and let A be a subset of Y. Prove that the topology A inherits as a subspace of Y is identical to the topology it inherits directly as a subspace of X.

Solution pending

Problem 2.11

Let A be a subset of a topological space X and $x \in X$ be an element. Then prove that the following statements are equivalent:

(i). The point $x \in \overline{A}$.

(ii). Every neighborhood of x intersects A.

(iii). Every basis element containing x intersects A.

Solution pending

Problem 2.12

Prove that $\overline{A} = A \cup A'$. Deduce that A is closed in X if and only if $A' \subseteq A$.

Solution

submitted by **Srijit Bhadra**

Solution: Let $x \in \bar{A}$. Now,

$$\begin{aligned} x \in \bar{A} &\iff \text{every neighbourhood of } x \text{ intersects } A \\ &\iff x \in A \text{ or } x \in A' \\ &\iff x \in A \cup A'. \end{aligned}$$

Since x was chosen arbitrarily, we can say $\bar{A} = A \cup A'$.

For the second part, assume A is closed. Then $\bar{A} = A \implies A \cup A' = A \implies A' \subseteq A$. On the other hand, if we assume $A' \subseteq A$, then we get $\bar{A} = A \cup A' = A$. Thus, in this case, we can conclude that A is closed. As both directions of the double implication hold, we have A is closed in X if and only if $A' \subseteq A$. □

Problem 2.13

Prove that $\operatorname{Int} A \cap \operatorname{Bd} A = \emptyset$, and $\overline{A} = \operatorname{Int} A \cup \operatorname{Bd} A$.

Solution

submitted by **Srijit Bhadra**

Solution: We know, $\operatorname{Int}(A) \cap \operatorname{Bd}(A) = \operatorname{Int}(A) \cap \bar{A} \cap \overline{X \setminus A}$. For the sake of contradiction, assume there is a point $x \in \operatorname{Int}(A) \cap \operatorname{Bd}(A) \implies x \in \operatorname{Int}(A) \cap \bar{A} \cap \overline{X \setminus A}$. As $x \in \overline{X \setminus A}$, every neighbourhood of x must intersect $X \setminus A$. Thus, $\operatorname{Int}(A) \cap \overline{X \setminus A} \neq \emptyset$. But this is impossible as $\operatorname{Int}(A) \subseteq A$ ($\Rightarrow\!\!\leftarrow$). Thus, $\operatorname{Int}(A) \cap \operatorname{Bd}(A) = \emptyset$.

For the second part, let $x \in \bar{A}$. Again, this happens if and only if every neighborhood of x intersects A. Now, either there exists a neighborhood U of x such that $x \in U \subseteq A$, or there is no such neighborhood. In the first case, $x \in U \subseteq A \implies x \in \operatorname{Int}(A)$ since $\operatorname{Int}(A)$ is the largest open set contained in A. In the second case, every neighborhood V of x intersects $X \setminus A \implies x \in \overline{X \setminus A} \implies x \in \operatorname{Bd}(A)$. Combining the two cases, $x \in \operatorname{Int}(A) \cup \operatorname{Bd}(A)$. Since $x \in \bar{A}$ was chosen arbitrarily, $\bar{A} \subseteq \operatorname{Int}(A) \cup \operatorname{Bd}(A)$.

For the other way inclusion, notice that $\text{Int}(A) \subseteq A \subseteq \bar{A}$. Also, $\text{Bd}(A) = \bar{A} \cap \overline{X \setminus A} \subseteq \bar{A}$. Therefore, $\text{Int}(A) \cup \text{Bd}(A) \subseteq \bar{A}$. Hence, we conclude that $\bar{A} = \text{Int}(A) \cup \text{Bd}(A)$. □

Problem 2.14

Prove that every order topology is Hausdorff.

Solution

submitted by **Srijit Bhadra**

Solution: Let $X \neq \emptyset$ and $\mathscr{T}$ be the order topology on X. Let $x_1, x_2 \in X$ such that $x_1 \neq x_2$. Consider the basic open set $(x_1, x_2) = \{x \in X : x_1 < x < x_2\} \in \mathscr{T}$. Now, there can be two cases:

Case 1. $(x_1, x_2) = \emptyset$. Observe that

$$(-\infty, x_2) = \bigcup_{a \in X,\ a < x_2} (a, x_2) \in \mathscr{T}.$$

Similarly,

$$(x_1, \infty) = \bigcup_{b \in X,\ b > x_1} (x_1, b) \in \mathscr{T}.$$

Also, $x_1 \in (-\infty, x_2)$ and $x_2 \in (x_1, \infty)$. Crucially, $(-\infty, x_2) \cap (x_1, \infty) = \emptyset$.

Case 2. $(x_1, x_2) \neq \emptyset$. Take any point $c \in (x_1, x_2)$. As we saw earlier, $(-\infty, c)$ and (c, ∞) are open in X. Note that $x_1 \in (-\infty, c)$, $x_2 \in (c, \infty)$, and $(-\infty, c) \cap (c, \infty) = \emptyset$.

Thus, in both cases, we found two disjoint open sets in X with one containing x_1 and the other containing x_2. As $x_1, x_2 \in X$ were chosen arbitrarily, we can say that $(X, \mathscr{T})$ is Hausdorff. Since $(X, \mathscr{T})$ was also chosen arbitrarily, we can say that every order topology is Hausdorff. □

Problem 2.15

Let X be a T_1 space. Then prove that for each pair x_1, x_2 of distinct points, there exist open sets U_1 and U_2 containing x_1 and x_2 respectively such that $x_1 \notin U_2$ and $x_2 \notin U_1$.

Solution

submitted by **Srijit Bhadra**

Solution: Let x_1, x_2 be a pair of distinct points in X. X being a T_1 space, singletons $\{x_1\}$ and $\{x_2\}$ are closed in X. Therefore, $X \setminus \{x_1\}$ and $X \setminus \{x_2\}$ are open in X. Now, $x_2 \in X \setminus \{x_1\}$ (because $x_1 \neq x_2$) but $x_2 \notin X \setminus \{x_2\}$. Similarly, $x_1 \in X \setminus \{x_2\}$ (because $x_1 \neq x_2$) but $x_1 \notin X \setminus \{x_1\}$. As $x_1, x_2 \in X$ were chosen arbitrarily, we can say that for each pair x_1, x_2 of distinct points, there exist open sets U_1 and U_2 containing x_1 and x_2, respectively, such that $x_1 \notin U_2$ and $x_2 \notin U_1$. □

Problem Set 3

Problem 3.1

Give an example of a T_1 space where a sequence converges to more than one point. Conclude that a T_1 space is not necessarily a Hausdorff space.

Solution *submitted by* **Arunodaya**

Consider the set $X = \mathbb{R}$ equipped with the cofinite topology

$$\tau = \{\emptyset\} \cup \{U \subseteq \mathbb{R} : \mathbb{R} \setminus U \text{ is finite}\},$$

that is, the open sets are the empty set or subsets of $\mathbb{R}$ whose complements are finite.

Now, we know that a topological space is T_1 if every singleton set is closed. Let $x \in \mathbb{R}$. Then $\mathbb{R} \setminus \{x\}$ is open because its complement $\{x\}$ is finite. Therefore, $\{x\}$ is closed. Since x is arbitrary, every singleton subset of $\mathbb{R}$ is closed. Hence $(\mathbb{R}, \tau)$ is a T_1 space.

Consider the sequence

$$x_n = n, \quad n \in \mathbb{N}.$$

$x_n \to x$ if for every open neighbourhood U of x, $\exists\, N \in \mathbb{N}$ such that $n \geq N \Rightarrow x_n \in U$.

Let $x \in \mathbb{R}$ be arbitrary. Let U be any open neighbourhood of x. By the definition of the cofinite topology, $\mathbb{R} \setminus U$ is finite. Since $x_n = n$, the sequence consists of distinct natural numbers. Consequently only finitely many terms of the sequence can belong to the finite set $\mathbb{R} \setminus U$. Therefore, $\exists\, N \in \mathbb{N}$ such that

$$n \geq N \Rightarrow n \notin \mathbb{R} \setminus U.$$

Thus, $n \geq N \Rightarrow x_n = n \in U$. So, by the definition of convergence,

$$x_n \to x \quad \forall\, x \in \mathbb{R}.$$

Now, the sequence $x_n = n$ converges to every real number

$\Rightarrow$ its limit is not unique

$\Rightarrow$ the space cannot be Hausdorff.

Thus $(\mathbb{R}, \tau)$ is T_1 but not T_2, i.e. $T_1 \not\Rightarrow T_2$.

Problem 3.2

Prove that the product topology $\mathcal{T}_{\text{prod}}$ on $\mathbb{R}^n$ (where each factor $\mathbb{R}$ is equipped with its standard Euclidean topology) coincides with the Euclidean topology $\mathcal{T}_{\text{Euc}}$ on $\mathbb{R}^n$ induced by the standard Euclidean metric.

Solution

submitted by **Arunodaya**

Let $U = U_1 \times U_2 \times \cdots \times U_n$ be a basic open set in the product topology. Take an arbitrary point $x = (x_1, x_2, x_3, x_4, \ldots, x_n) \in U$. Since each U_i is open in $\mathbb{R}$, $\exists\, \varepsilon_i > 0$ such that

$$(x_i - \varepsilon_i,\, x_i + \varepsilon_i) \subseteq U_i.$$

Choose $\varepsilon = \min\{\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n\}$. Since there are finitely many positive numbers, $\varepsilon > 0$.

Now, consider any point $y = (y_1, y_2, \ldots, y_n) \in B(x, \varepsilon)$. Then $d(x, y) < \varepsilon$. Thus,

$$\sqrt{\sum_{i=1}^{n}(x_i - y_i)^2} < \varepsilon \;\Rightarrow\; \sum_{i=1}^{n}(x_i - y_i)^2 < \varepsilon^2.$$

Since every term in the sum is non-negative, $(x_i - y_i)^2 < \varepsilon^2$ for every $i = 1, 2, \ldots, n$. Consequently $|x_i - y_i| < \varepsilon \leq \varepsilon_i$; thus $y_i \in (x_i - \varepsilon_i,\, x_i + \varepsilon_i) \subseteq U_i$. This holds for every i. Hence

$$y \in U_1 \times U_2 \times \cdots \times U_n \;\Rightarrow\; B(x, \varepsilon) \subseteq U.$$

Since $x \in U$ was arbitrary, every point of U has a Euclidean open ball contained in U. Hence U is Euclidean open.

$$\underline{\mathcal{T}_{\text{prod}} \subseteq \mathcal{T}_{\text{Euc}}}$$

Let V be an arbitrary Euclidean open subset of $\mathbb{R}^n$. Take any point $x = (x_1, x_2, \ldots, x_n) \in V$. Since V is Euclidean open, $\exists\, \delta > 0$ such that $B(x, \delta) \subseteq V$. Now consider the product of open intervals

$$W = \prod_{i=1}^{n}\Big(x_i - \frac{\delta}{\sqrt{n}},\; x_i + \frac{\delta}{\sqrt{n}}\Big).$$

We will show that $W \subseteq B(x, \delta)$. Take any $y = (y_1, y_2, y_3, y_4, \ldots, y_n) \in W$. By definition of W,

$$|x_i - y_i| < \delta/\sqrt{n} \quad \forall\, i = 1, 2, 3, 4, \ldots, n.$$

Therefore

$$d(x, y) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2} < \sqrt{\sum_{i=1}^{n}\frac{\delta^2}{n}} = \sqrt{\frac{n\delta^2}{n}} = \delta.$$

Thus $y \in B(x, \delta)$. Consequently $W \subseteq B(x, \delta) \subseteq V$.

Notice that W is a basic open set in the product topology because it is a product of open intervals in $\mathbb{R}$. Hence, every point $x \in V$ has a basic product open neighbourhood contained in V. Therefore, V is open in the product topology. So, $\underline{\mathcal{T}_{\text{Euc}} \subseteq \mathcal{T}_{\text{prod}}}$.

So $\mathcal{T}_{\text{prod}} = \mathcal{T}_{\text{Euc}}$.

Problem 3.3

Let $\{X_\alpha\}_{\alpha\in\Lambda}$ be a family of topological spaces, and let $\phi\colon \Lambda \to \Lambda$ be a bijection. Prove that the product spaces $X = \prod_{\alpha\in\Lambda} X_\alpha$ and $X_\phi = \prod_{\alpha\in\Lambda} X_{\phi(\alpha)}$, both equipped with their respective product topologies, are homeomorphic.

Solution

submitted by **Arunodaya**

We define $f\colon X \to X_\phi$ by

$$f\big((x_\alpha)_{\alpha\in A}\big) = \big(x_{\phi(\alpha)}\big)_{\alpha\in A}.$$

Since ϕ is a bijection, every coordinate in the second product corresponds to exactly one coordinate in the first product. Since ϕ is bijective, it has an inverse

$$\phi^{-1}\colon B \to A.$$

Define $g\colon X_\phi \to X$ by

$$g\big((y_\beta)_{\beta\in B}\big) = \big(y_{\phi^{-1}(\beta)}\big)_{\beta\in B}.$$

Then for every $x = (x_\alpha) \in X$, $g(f(x)) = x$. Similarly for every $y \in X_\phi$, $f(g(y)) = y$. Thus $g = f^{-1}$. Hence f is bijective.

The product topology has a subbasis consisting of sets of the form $\pi_\alpha^{-1}(U_\alpha)$, where U_α is open in X_α and $\pi_\alpha\colon X \to X_\alpha$ is the coordinate projection.

Let U_α be open in $X_{\phi(\alpha)}$. Consider the subbasic open set in X_ϕ

$$V = \{y \in X_\phi : y_\alpha \in U_\alpha\}.$$

$$\begin{aligned} f^{-1}(V) &= \{x \in X : f(x)_\alpha \in U_\alpha\} \\ &= \{x \in X : x_\alpha \in U_\alpha\} \\ &= \pi_\alpha^{-1}(U_\alpha). \end{aligned}$$

Since U_α is open and π_α is continuous, the set is open in $X \Rightarrow$ the inverse image of every subbasic open set is open. Consequently, f is continuous.

By the same argument, for every subbasic open set

$$W = \{x \in X : x_\alpha \in U_\alpha\}$$

we have

$$(f^{-1})^{-1}(W) = \{y \in X_\phi : y_{\phi^{-1}(\alpha)} \in U_\alpha\};$$

this is a subbasic open set in X_ϕ. Hence f^{-1} is continuous.

We proved that f is bijective, continuous and has a continuous inverse; therefore $X \cong X_\phi$.

Problem 3.4

Let $\{X_\alpha\}_{\alpha\in\Lambda}$ be a family of topological spaces, and for each $\alpha \in \Lambda$, let $Y_\alpha \subseteq X_\alpha$ be equipped with the subspace topology. Prove that the product topology on $Y = \prod_{\alpha\in\Lambda} Y_\alpha$ coincides with the subspace topology on Y induced by the product topology on $X = \prod_{\alpha\in\Lambda} X_\alpha$.

Solution

submitted by **Arunodaya**

Let

- τ_1 denote the product topology on Y,
- τ_2 denote the subspace topology on Y inherited from X.

We shall prove $\tau_1 = \tau_2$.

Let B be a basic open set in X. Then

$$B = \bigcap_{\alpha\in F} \pi_\alpha^{-1}(U_\alpha),$$

where each U_α is open in X_α. By the definition of the subspace topology, a basic open set in Y is $B \cap Y$. Now,

$$\begin{aligned} B \cap Y &= Y \cap \bigcap_{\alpha\in F} \pi_\alpha^{-1}(U_\alpha) \\ &= \bigcap_{\alpha\in F} \left(Y \cap \pi_\alpha^{-1}(U_\alpha)\right) \\ &= \bigcap_{\alpha\in F} q_\alpha^{-1}(U_\alpha \cap Y_\alpha). \end{aligned}$$

Since U_α is open in X_α, the set $U_\alpha \cap Y_\alpha$ is open in Y_α by the subspace topology. Therefore, $B \cap Y$ is open in the product topology on Y. Hence $\underline{\tau_2 \subseteq \tau_1}$.

Let C be a basic open set in the product topology on Y. Then

$$C = \bigcap_{\alpha\in F} q_\alpha^{-1}(V_\alpha),$$

where V_α is open in Y_α. Since Y_α has the subspace topology inherited from X_α, $\exists$ an open set U_α in X_α such that $V_\alpha = U_\alpha \cap Y_\alpha$. Therefore,

$$\begin{aligned} C &= \bigcap_{\alpha\in F} q_\alpha^{-1}(U_\alpha \cap Y_\alpha) \\ &= Y \cap \bigcap_{\alpha\in F} \pi_\alpha^{-1}(U_\alpha). \end{aligned}$$

But $\bigcap_{\alpha\in F} \pi_\alpha^{-1}(U_\alpha)$ is open in X because it is a basic open set in the product topology on X. Thus, C is the intersection of Y with an open set in X. Hence C is open in the subspace topology on Y. Therefore $\underline{\tau_1 \subseteq \tau_2}$.

So, $\underline{\tau_1 = \tau_2}$.

Problem 3.5

Let $\{X_\alpha\}_{\alpha\in\Lambda}$ be an indexed collection of topological spaces, and let $X = \prod_{\alpha\in\Lambda} X_\alpha$ be equipped with the product topology $\mathcal{T}_{\text{prod}}$. For any fixed index $\beta \in \Lambda$, consider the β-th coordinate projection map

$$p_\beta \colon \prod_{\alpha\in\Lambda} X_\alpha \longrightarrow X_\beta, \qquad p_\beta\big((x_\alpha)_{\alpha\in\Lambda}\big) = x_\beta.$$

Prove that p_β is an open map. Is p_β necessarily a closed map?

Solution

submitted by **Arunodaya**

Part (i). By the definition of the product topology, every basic open set has the form

$$B = \bigcap_{\alpha\in F} \pi_\alpha^{-1}(U_\alpha),$$

where

- $F \subseteq A$ is finite,
- each U_α is open in X_α.

Case 1: $\beta \in F$. Then $B = \bigcap_{\alpha\in F} \pi_\alpha^{-1}(U_\alpha)$, and the image of B under p_β is

$$p_\beta(B) = U_\beta,$$

provided the other coordinate restrictions are non-empty. Indeed, every element of B has its β-coordinate in U_β. Conversely, any element of U_β can be extended to a point of B if all the other coordinate restrictions are non-empty. Since U_β is open in X_β, the image is open. If another coordinate restriction is empty, then $B = \emptyset$, whose image is also open.

Case 2: $\beta \notin F$. In this case, there is no restriction on the β-coordinate:

$$p_\beta(B) = X_\beta,$$

provided $B \neq \emptyset$. Since X_β is open in itself, the image is open. If $B = \emptyset$, its image is open again.

Now, $U = \bigcup_{\lambda\in\Lambda} B_\lambda$. Using the fact that images preserve arbitrary unions, we get

$$p_\beta(U) = p_\beta\Big(\bigcup_{\lambda\in\Lambda} B_\lambda\Big) = \bigcup_{\lambda\in\Lambda} p_\beta(B_\lambda).$$

From the above step, $p_\beta(B_\lambda)$ is open in X_β. An arbitrary union of open sets is open; thus, $p_\beta(U)$ is open in X_β.

Part (ii). No. Take $X_1 = \mathbb{R}$, $X_2 = \mathbb{R}$ with their usual Euclidean topologies. Then $X = X_1 \times X_2 = \mathbb{R}^2$. Consider the coordinate projection $p_1 \colon \mathbb{R}^2 \to \mathbb{R}$ defined by $p_1(x, y) = x$. Consider

$$F = \{(x, y) \in \mathbb{R}^2 : xy = 1\}.$$

Define $h \colon \mathbb{R}^2 \to \mathbb{R}$ by $h(x, y) = xy$. Since multiplication is continuous, h is continuous. Also $F = h^{-1}(\{1\})$. Since $\{1\}$ is closed in $\mathbb{R}$ and the inverse image of a closed set under a continuous function is closed, we say F is closed in $\mathbb{R}^2$.

We have

$$p_1(F) = \{x \in \mathbb{R} \mid \exists\, y \in \mathbb{R} \text{ such that } xy = 1\} = \mathbb{R} \setminus \{0\}.$$

Indeed, if $x \neq 0$ we can choose $y = 1/x$. But if $x = 0$, no such y exists. Therefore $p_1(F) = \mathbb{R} \setminus \{0\}$.

The set $\mathbb{R} \setminus \{0\} = (-\infty, 0) \cup (0, \infty)$ is not closed in $\mathbb{R}$ because its complement $\{0\}$ is not open in $\mathbb{R}$. Thus, we found a closed set $F \subseteq \mathbb{R}^2$ whose image under p_1 is not closed. Consequently, p_1 is not closed.

So, every coordinate projection is an open map. A coordinate projection need not be a closed map.

Problem 3.6

Let L be a line in the plane $\mathbb{R}^2$, and $\mathbb{R}_\ell$ be the lower limit topology on $\mathbb{R}$. Discuss the topology L inherits as a subspace of $\mathbb{R}_\ell \times \mathbb{R}$ and as a subspace of $\mathbb{R}_\ell \times \mathbb{R}_\ell$.

Solution

submitted by **Pula Karan**

We analyze the subspace topology on L under both product topologies.

Part 1: Subspace Topology inherited from $\mathbb{R}_l \times \mathbb{R}$. A basis for $\mathbb{R}_l \times \mathbb{R}$ consists of rectangles of the form $[a, b) \times (c, d)$ with $a < b$ and $c < d$.

(a) **Vertical Line ($L : x = k$):** For any $(k, y_0) \in L$, a basic open neighborhood in $\mathbb{R}_l \times \mathbb{R}$ is $U = [k, k + \epsilon) \times (y_0 - \epsilon, y_0 + \epsilon)$. Intersecting U with L yields:

$$U \cap L = \{k\} \times (y_0 - \epsilon, y_0 + \epsilon)$$

This traces open intervals along the y-axis. Thus, L inherits the **standard Euclidean topology** ($\mathbb{R}$).

(b) **Horizontal Line ($L : y = k$):** For any $(x_0, k) \in L$, a basic open neighborhood in $\mathbb{R}_l \times \mathbb{R}$ is $U = [x_0, x_0 + \epsilon) \times (k - \epsilon, k + \epsilon)$. Intersecting U with L yields:

$$U \cap L = [x_0, x_0 + \epsilon) \times \{k\}$$

This traces right-half-open intervals along the x-axis. Thus, L inherits the **lower limit topology** ($\mathbb{R}_l$).

(c) **Non-Vertical Line ($L : y = mx + k$, $m \neq 0$):** For any $(x_0, y_0) \in L$, consider basic open sets $U = [x_0, x_0 + \epsilon) \times (y_0 - \delta, y_0 + \delta)$.

- If $m > 0$, as x increases from x_0, y increases from y_0. For sufficiently small $\delta > 0$, $U \cap L$ isolates points corresponding to $x \in [x_0, x_0 + \delta/m)$.
- If $m < 0$, as x increases from x_0, y decreases from y_0. Intersecting $[x_0, x_0 + \epsilon)$ with $(y_0 - \delta, y_0 + \delta)$ again isolates points corresponding to $x \in [x_0, x_0 + \delta/|m|)$.

In both cases, basic open sets induce half-open intervals $[x_0, x_0 + r)$ along L. Thus, L inherits the **lower limit topology** ($\mathbb{R}_l$).

Part 2: Subspace Topology inherited from $\mathbb{R}_l \times \mathbb{R}_l$ (Sorgenfrey Plane). A basis for $\mathbb{R}_l \times \mathbb{R}_l$ consists of half-open rectangles of the form $[a, b) \times [c, d)$.

(a) **Vertical ($x = k$), Horizontal ($y = k$), and Positive Slope ($y = mx + k$, $m > 0$):** For any $(x_0, y_0) \in L$, intersecting $[x_0, x_0 + \epsilon) \times [y_0, y_0 + \epsilon')$ with L isolates points corresponding to $[x_0, x_0 + \delta)$ where $\delta = \min(\epsilon, \epsilon'/m)$ for $m > 0$ (or ϵ if vertical/horizontal). Thus, L inherits the **lower limit topology** ($\mathbb{R}_l$).

(b) **Oblique Line with Negative Slope ($y = mx + k$, $m < 0$):** Let $(x_0, y_0) \in L$ and consider $U = [x_0, x_0 + \epsilon) \times [y_0, y_0 + \epsilon)$. For any $x > x_0$, $y = mx + k < mx_0 + k = y_0$, so $(x, y) \notin U$. For $x < x_0$, $x \notin [x_0, x_0 + \epsilon)$, so $(x, y) \notin U$. Therefore, $U \cap L = \{(x_0, y_0)\}$. Since every singleton is open, L inherits the **discrete topology**.

Problem 3.7

Compare (i) the ordered square I_o^2, (ii) the topology I^2 inherits as a subspace of the product topology $\mathbb{R}^2$, and (iii) the product topology on $I \times I$.

Solution

submitted by **Pula Karan**

Let $I = [0, 1]$. We compare the three topologies on $I^2 = [0, 1] \times [0, 1]$:

1. **Topologies (ii) and (iii) are identical ($\mathcal{T}_{\text{sub}} = \mathcal{T}_{\text{prod}}$):**
   - Topology (ii) is the subspace topology inherited from the Euclidean product topology on $\mathbb{R}^2$. Basis elements are of the form $(U \times V) \cap I^2$, where $U, V \subseteq \mathbb{R}$ are open intervals.
   - Topology (iii) is the product topology on $I \times I$, where each factor I carries the subspace topology from $\mathbb{R}$. Basis elements are of the form $(U \cap I) \times (V \cap I) = (U \times V) \cap I^2$.

   Since their bases coincide, Topologies (ii) and (iii) are identical.
2. **Topology (i) (I_o^2) is strictly finer than Topologies (ii) and (iii):** The ordered square I_o^2 is equipped with the dictionary order topology $\leq_{\text{dic}}$.
   - *Euclidean $\subseteq$ Dictionary Order*: Any standard Euclidean basic open set $U = (a, b) \times (c, d) \subset (0, 1)^2$ can be written as:

     $$U = \bigcup_{x_0 \in (a,b)} \Big(\{x_0\} \times (c, d) \Big) = \bigcup_{x_0 \in (a,b)} \Big((x_0, c), (x_0, d) \Big)_{\text{dic}}$$

     Since each vertical interval $\{x_0\} \times (c, d)$ is open in the dictionary order topology, U is open in I_o^2.
   - *Strict Containment*: Sets of the form $\{x_0\} \times (c, d)$ for $0 < c < d < 1$ are open in I_o^2. However, they contain no 2D Euclidean open ball, so they are not open in the standard topology on I^2.

Conclusion: Topology (ii) = Topology (iii) $\subsetneq$ Topology (i).

Problem 3.8

Let X be a topological space, and $\Delta = \{(x, x) \in X \times X \mid x \in X\}$ be the diagonal of $X \times X$, equipped with the subspace topology inherited from the product topology on $X \times X$.

(i) Prove that X is Hausdorff if and only if the diagonal Δ is a closed subset of $X \times X$.

(ii) Prove that the diagonal map $d\colon X \to \Delta$ defined by $d(x) = (x, x)$ is a homeomorphism.

Solution *submitted by* **Pula Karan**

Part (i).

Proof. ($\implies$) Suppose X is Hausdorff. We show that $(X \times X) \setminus \Delta$ is open. Let $(x, y) \in (X \times X) \setminus \Delta$, which means $x \neq y$. Since X is Hausdorff, there exist disjoint open sets $U, V \subseteq X$ such that $x \in U$, $y \in V$, and $U \cap V = \varnothing$. Then $U \times V$ is an open neighborhood of (x, y) in $X \times X$. If there were some $(z, z) \in (U \times V) \cap \Delta$, then $z \in U \cap V = \varnothing$, a contradiction. Hence $(U \times V) \cap \Delta = \varnothing$, so $U \times V \subseteq (X \times X) \setminus \Delta$. Thus $(X \times X) \setminus \Delta$ is open, so Δ is closed.

($\impliedby$) Suppose Δ is closed in $X \times X$, so $(X \times X) \setminus \Delta$ is open. Let $x, y \in X$ with $x \neq y$. Then $(x, y) \in (X \times X) \setminus \Delta$. By definition of the product topology, there exists a basic open set $U \times V \subseteq X \times X$ such that $(x, y) \in U \times V \subseteq (X \times X) \setminus \Delta$. This implies $x \in U$ and $y \in V$. If there existed $z \in U \cap V$, then $(z, z) \in (U \times V) \cap \Delta$, contradicting $U \times V \subseteq (X \times X) \setminus \Delta$. Therefore $U \cap V = \varnothing$, so X is Hausdorff. □

Part (ii).

Proof. We check the three properties of a homeomorphism for $d\colon X \to \Delta$:

1. **Bijectivity:**
   - *Injective*: $d(x_1) = d(x_2) \implies (x_1, x_1) = (x_2, x_2) \implies x_1 = x_2$.
   - *Surjective*: Any element in Δ is of the form $(x, x) = d(x)$ for some $x \in X$.
2. **Continuity of d:** The component functions of d are $d_1 = \mathrm{id}_X$ and $d_2 = \mathrm{id}_X$. Since the identity map $\mathrm{id}_X\colon X \to X$ is continuous, $d\colon X \to X \times X$ is continuous by the characteristic property of product spaces. Restricting the codomain to the subspace Δ preserves continuity.
3. **Continuity of d^{-1}:** The inverse map $d^{-1}\colon \Delta \to X$ is given by $d^{-1}(x, x) = x$. This is the restriction of the first coordinate projection $p_1\colon X \times X \to X$ to Δ. Since projection maps are continuous, its restriction $d^{-1} = p_1|_\Delta$ is continuous.

Thus, $d\colon X \to \Delta$ is a homeomorphism. □

Problem 3.9

Let X and Y be topological spaces, and let $f, g\colon X \to Y$ be continuous maps. Assume that Y is a Hausdorff space. Prove the following statements:

(i) The coincidence set $E = \{x \in X \mid f(x) = g(x)\}$ is closed in X.

(ii) If D is a dense subset of X and $f|_D = g|_D$, then $f = g$ on all of X.

(iii) The graph of f, defined as $\Gamma_f = \{(x, f(x)) \in X \times Y \mid x \in X\}$, is closed in the product space $X \times Y$.

(iv) Assume that $f\colon X \to Y$ is a continuous, open, and surjective map. Then the fiber relation set $R = \{(x_1, x_2) \in X \times X \mid f(x_1) = f(x_2)\}$ is closed in $X \times X$ if and only if Y is Hausdorff.

Solution *submitted by* **Pula Karan**

Part (i).

Proof. Define $h\colon X \to Y \times Y$ by $h(x) = (f(x), g(x))$. Since f and g are continuous, h is continuous. Because Y is Hausdorff, the diagonal $\Delta_Y = \{(y, y) \in Y \times Y\}$ is closed in $Y \times Y$ by Problem 3.8(i). Notice that:

$$x \in E \iff f(x) = g(x) \iff (f(x), g(x)) \in \Delta_Y \iff x \in h^{-1}(\Delta_Y)$$

Thus $E = h^{-1}(\Delta_Y)$. Since Δ_Y is closed and h is continuous, E is closed in X. □

Part (ii).

Proof. Consider the coincidence set $E = \{x \in X \mid f(x) = g(x)\}$. By part (i), E is closed in X. Since $f|_D = g|_D$, we have $D \subseteq E$. Taking closures gives $\overline{D} \subseteq \overline{E} = E$. Since D is dense in X, $\overline{D} = X$. Therefore $X \subseteq E$, which means $E = X$, so $f(x) = g(x)$ for all $x \in X$. □

Part (iii).

Proof. Define $F\colon X \times Y \to Y \times Y$ by $F(x, y) = (f(x), y)$. F is continuous because its component maps $f \circ p_1$ and p_2 are continuous. Since Y is Hausdorff, Δ_Y is closed in $Y \times Y$. Notice that:

$$(x, y) \in \Gamma_f \iff y = f(x) \iff (f(x), y) \in \Delta_Y \iff F(x, y) \in \Delta_Y$$

Thus $\Gamma_f = F^{-1}(\Delta_Y)$. Since F is continuous and Δ_Y is closed, Γ_f is closed in $X \times Y$. □

Part (iv).

Proof. Consider the product map $f \times f\colon X \times X \to Y \times Y$ defined by $(f \times f)(x_1, x_2) = (f(x_1), f(x_2))$. Since f is continuous, open, and surjective, $f \times f$ is also continuous, open, and surjective. Notice that $R = (f \times f)^{-1}(\Delta_Y)$.

($\Longleftarrow$) If Y is Hausdorff, Δ_Y is closed in $Y \times Y$. By continuity of $f \times f$, $R = (f \times f)^{-1}(\Delta_Y)$ is closed in $X \times X$.

($\implies$) Suppose R is closed in $X \times X$. Since $f \times f$ is surjective, $(f \times f)(R) = \Delta_Y$. Taking complements yields:

$$(Y \times Y) \setminus \Delta_Y = (f \times f)((X \times X) \setminus R)$$

Since R is closed, $(X \times X) \setminus R$ is open. Because $f \times f$ is an open map, $(Y \times Y) \setminus \Delta_Y$ is open in $Y \times Y$, which means Δ_Y is closed in $Y \times Y$. By Problem 3.8(i), Y is Hausdorff. □

Problem 3.10

Let X be a topological space and $A \subset X$ be dense in X. Let Y be a Hausdorff space and $f\colon A \to Y$ be a continuous map. Show that if f may be extended to a continuous map $g\colon X \to Y$, then g is uniquely determined by f.

Solution

submitted by **Pula Karan**

Proof. Suppose $g_1, g_2\colon X \to Y$ are two continuous extensions of $f\colon A \to Y$, so $g_1|_A = f$ and $g_2|_A = f$.

1. Since $g_1|_A = g_2|_A = f$, the maps g_1 and g_2 agree on the subset $A \subseteq X$.
2. A is dense in X, and Y is a Hausdorff space.
3. By **Problem 3.9(ii)**, if two continuous functions from X into a Hausdorff space Y agree on a dense subset $A \subseteq X$, they must agree on all of X.

Therefore, $g_1(x) = g_2(x)$ for all $x \in X$, proving that the continuous extension g is uniquely determined by f. □

Problem Set 4

Problem 4.1

Let X and Y be two topological spaces; and $a \in X$, $b \in Y$ be two given points. Prove that the maps $f\colon X \to X \times Y$ and $g\colon Y \to X \times Y$ given by

$$f(x) = (x, b) \quad \text{and} \quad g(y) = (a, y)$$

are embeddings.

Solution

submitted by **Dhrubojoti**

Proof. Recall that a map $h\colon Z \to W$ is an embedding if it is a homeomorphism from Z onto $h(Z)$, equipped with the subspace topology.

1. The map $f\colon X \to X \times Y$, $f(x) = (x, b)$. *Step 1: f is continuous.* Let $U \times V$ be an arbitrary basic open set in $X \times Y$, where $U \subseteq X$ and $V \subseteq Y$ are open. Then

$$\begin{aligned} f^{-1}(U \times V) &= \{x \in X : f(x) \in U \times V\} \\ &= \{x \in X : (x, b) \in U \times V\} \\ &= \begin{cases} U, & b \in V, \\ \emptyset, & b \notin V. \end{cases} \end{aligned}$$

In either case, $f^{-1}(U \times V)$ is open in X. Hence f is continuous.

Step 2: f is injective. Suppose that $f(x_1) = f(x_2)$. Then

$$(x_1, b) = (x_2, b).$$

Therefore, $x_1 = x_2$, so f is injective.

Step 3: The inverse map $f^{-1}\colon f(X) \to X$ is continuous. Observe that

$$f(X) = X \times \{b\}.$$

Consider the first projection map

$$\pi_X\colon X \times Y \to X, \qquad \pi_X(x, y) = x.$$

Since projection maps are continuous, its restriction

$$\pi_X|_{f(X)}\colon f(X) \to X$$

is continuous. But

$$\pi_X(f(x)) = \pi_X(x, b) = x.$$

Thus

$$f^{-1} = \pi_X|_{f(X)}.$$

Hence f^{-1} is continuous.

Therefore, f is a homeomorphism from X onto $f(X)$, and consequently f is an embedding.

2. The map $g\colon Y \to X \times Y$, $g(y) = (a, y)$. *Step 1: g is continuous.* Let $U \times V$ be an arbitrary basic open set in $X \times Y$. Then

$$\begin{aligned} g^{-1}(U \times V) &= \{y \in Y : g(y) \in U \times V\} \\ &= \{y \in Y : (a, y) \in U \times V\} \\ &= \begin{cases} V, & a \in U, \\ \emptyset, & a \notin U. \end{cases} \end{aligned}$$

In either case, $g^{-1}(U \times V)$ is open in Y. Hence g is continuous.

Step 2: g is injective. Suppose that $g(y_1) = g(y_2)$. Then

$$(a, y_1) = (a, y_2).$$

Therefore, $y_1 = y_2$, so g is injective.

Step 3: The inverse map $g^{-1}\colon g(Y) \to Y$ is continuous. Observe that

$$g(Y) = \{a\} \times Y.$$

Consider the second projection map

$$\pi_Y\colon X \times Y \to Y, \qquad \pi_Y(x, y) = y.$$

Since projection maps are continuous, its restriction

$$\pi_Y|_{g(Y)}\colon g(Y) \to Y$$

is continuous. But

$$\pi_Y(g(y)) = \pi_Y(a, y) = y.$$

Thus

$$g^{-1} = \pi_Y|_{g(Y)}.$$

Hence g^{-1} is continuous.

Therefore, g is a homeomorphism from Y onto $g(Y)$, and consequently g is an embedding.

Hence both f and g are embeddings.

Problem 4.2

Suppose that $f, g\colon X \to \mathbb{R}$ are two continuous maps.

(i). Prove that the set $\{x \in X \mid f(x) \leq g(x)\}$ is closed in X.

(ii). Prove that the map $h\colon X \to \mathbb{R}$ given by $h(x) = \min\{f(x), g(x)\}$ is continuous.

Solution *submitted by* **Dhrubojoti**

Proof.

(i) The set $\{x \in X \mid f(x) \leq g(x)\}$ is closed in X. Let

$$A = \{x \in X \mid f(x) \leq g(x)\}.$$

We know that f and g are continuous. Since the subtraction map

$$\varphi\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}, \qquad \varphi(u, v) = u - v$$

is continuous, the map

$$F\colon X \to \mathbb{R}, \qquad F(x) = f(x) - g(x)$$

is continuous. Now observe that

$$\begin{aligned} A &= \{x \in X \mid f(x) \leq g(x)\} \\ &= \{x \in X \mid f(x) - g(x) \leq 0\} \\ &= F^{-1}((-\infty, 0]). \end{aligned}$$

Since $(-\infty, 0]$ is closed in $\mathbb{R}$ and F is continuous, its inverse image is closed in X. Therefore,

$$\boxed{\{x \in X \mid f(x) \leq g(x)\} \text{ is closed in } X.}$$

(ii) The map $h\colon X \to \mathbb{R}$ given by $h(x) = \min\{f(x), g(x)\}$ is continuous. We use the identity

$$\min\{a, b\} = \frac{a + b - |a - b|}{2}, \qquad a, b \in \mathbb{R}.$$

Therefore, for every $x \in X$,

$$h(x) = \frac{f(x) + g(x) - |f(x) - g(x)|}{2}.$$

Since f and g are continuous, the maps

$$f + g\colon X \to \mathbb{R} \quad \text{and} \quad f - g\colon X \to \mathbb{R}$$

are continuous. Also, the absolute value map

$$|\cdot|\colon \mathbb{R} \to \mathbb{R}, \qquad t \mapsto |t|$$

is continuous. Hence the composition

$$x \mapsto |f(x) - g(x)|$$

is continuous. Consequently,

$$h(x) = \frac{f(x) + g(x) - |f(x) - g(x)|}{2}$$

is continuous, since sums, differences and scalar multiples of continuous real-valued functions are continuous. Thus,

$$\boxed{h\colon X \to \mathbb{R} \text{ is continuous.}}$$

Problem 4.3

Let $\{A_\alpha\}$ be a collection of subsets of X such that $X = \bigcup_\alpha A_\alpha$. Suppose that $f\colon X \to Y$ is a map such that the restriction map $f|_{A_\alpha}\colon A_\alpha \to Y$ is continuous for each α.

(i). Prove that if the collection $\{A_\alpha\}$ is finite and each set A_α is closed, then f is continuous.

(ii). Find an example where the collection $\{A_\alpha\}$ is countable and each set A_α is closed, but f is not continuous.

(iii). An indexed family of sets $\{A_\alpha\}$ is said to be *locally finite* if each point x has a neighborhood that intersects A_α for only finitely many values of α. Prove that if the family $\{A_\alpha\}$ is locally finite and each set A_α is closed, then f is continuous.

Solution

submitted by **Dhrubojoti**

Recall. A map $f\colon X \to Y$ is continuous if and only if the preimage of every closed subset of Y is closed in X. Also, if A is a closed subset of X, then every relatively closed subset of A is closed in X.

(i) The collection $\{A_\alpha\}$ is finite and each A_α is closed.

Proof. Suppose

$$X = A_1 \cup A_2 \cup \cdots \cup A_n,$$

where each A_i is closed in X. We are given that the restriction

$$f|_{A_i}\colon A_i \to Y$$

is continuous for every $i = 1, 2, \ldots, n$. We need to prove that $f\colon X \to Y$ is continuous.

Let C be an arbitrary closed subset of Y. Consider

$$f^{-1}(C) = \{x \in X : f(x) \in C\}.$$

Since $X = \bigcup_{i=1}^n A_i$, we can write

$$f^{-1}(C) = \bigcup_{i=1}^{n} (f^{-1}(C) \cap A_i).$$

Now observe that

$$f^{-1}(C) \cap A_i = (f|_{A_i})^{-1}(C).$$

Since $f|_{A_i}$ is continuous and C is closed in Y, $(f|_{A_i})^{-1}(C)$ is closed in A_i. But A_i itself is closed in X. Therefore, a relatively closed subset of A_i is also closed in X. Thus, $f^{-1}(C) \cap A_i$ is closed in X for every i. Consequently,

$$f^{-1}(C) = \bigcup_{i=1}^{n} (f^{-1}(C) \cap A_i)$$

is a finite union of closed subsets of X. Hence, $f^{-1}(C)$ is closed in X. Since C was an arbitrary closed subset of Y, we conclude that

$$\boxed{f\colon X \to Y \text{ is continuous.}}$$

□

(ii) Find a countable collection of closed sets covering X for which f is not continuous. Example. Take

$$X = \mathbb{R}, \qquad Y = \mathbb{R}.$$

Define

$$A_0 = \{0\}$$

and, for $n \in \mathbb{N}$,

$$A_n = \left[-n, -\frac{1}{n}\right] \cup \left[\frac{1}{n}, n\right].$$

Step 1. Show that each A_n is closed. Clearly, $A_0 = \{0\}$ is closed in $\mathbb{R}$. Also, each A_n is a finite union of closed intervals. Hence, every A_n is closed in $\mathbb{R}$.

Step 2. Show that the collection is countable and covers $\mathbb{R}$. The collection

$$\{A_0, A_1, A_2, \dots\}$$

is countable. Moreover, every nonzero real number belongs to some A_n. Indeed, given $x \neq 0$, choose n sufficiently large such that

$$n \geq |x| \quad \text{and} \quad \frac{1}{n} \leq |x|.$$

Then $x \in A_n$. Since $0 \in A_0$, we obtain

$$\boxed{\mathbb{R} = \bigcup_{n=0}^{\infty} A_n.}$$

Step 3. Define the map f. Define $f\colon \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} 1, & x = 0, \\ 0, & x \neq 0. \end{cases}$$

We now show that the restriction of f to every A_n is continuous.

- On $A_0 = \{0\}$, the restriction is the constant function $f(0) = 1$, which is continuous.
- On every A_n, $n \geq 1$, we have $0 \notin A_n$. Therefore, $f|_{A_n}$ is the constant zero function, which is continuous.

Thus, every restriction $f|_{A_n}\colon A_n \to \mathbb{R}$ is continuous.

Step 4. Show that f is not continuous. Consider the open set

$$U = \left(\frac{1}{2}, \frac{3}{2}\right) \subseteq \mathbb{R}.$$

We have

$$f^{-1}(U) = \{0\}.$$

But $\{0\}$ is not open in $\mathbb{R}$. Therefore, f is not continuous.

Hence, we have constructed a countable closed cover of $\mathbb{R}$ such that every restriction is continuous, but the original map is not continuous. Thus,

$$\boxed{\text{The conclusion of (i) need not hold for a countable cover.}}$$

□

(iii) Prove the result when $\{A_\alpha\}$ is locally finite and each A_α is closed.

Proof. We are given that:

1. $X = \bigcup_\alpha A_\alpha$.
2. Every A_α is closed in X.
3. The family $\{A_\alpha\}$ is locally finite.
4. Every restriction $f|_{A_\alpha} : A_\alpha \to Y$ is continuous.

We need to prove that $f : X \to Y$ is continuous. We will use the following important result.

Lemma. *The union of a locally finite family of closed subsets of a topological space is closed.*

Proof of the lemma. Let $\{F_\alpha\}$ be a locally finite family of closed subsets of X. We want to show that

$$F = \bigcup_\alpha F_\alpha$$

is closed in X. Take an arbitrary point $x \in X \setminus F$. Since the family is locally finite, there exists an open neighbourhood U of x that intersects only finitely many members of the family. Suppose these members are

$$F_{\alpha_1}, F_{\alpha_2}, \ldots, F_{\alpha_n}.$$

Now, $x \notin F$, so in particular,

$$x \notin F_{\alpha_i}, \qquad i = 1, \ldots, n.$$

Since each F_{α_i} is closed, its complement is open. Thus, $X \setminus F_{\alpha_i}$ is an open neighbourhood of x. Consider

$$V = U \cap \bigcap_{i=1}^{n} (X \setminus F_{\alpha_i}).$$

This is an open neighbourhood of x, being a finite intersection of open sets. We claim that

$$V \cap F = \emptyset.$$

Indeed:

- For the finitely many sets $F_{\alpha_1}, \ldots, F_{\alpha_n}$, the set V does not intersect any of them.
- For every other index α, U, and hence V, does not intersect F_α, by the choice of U.

Therefore, V does not intersect their union F. So every point in $X \setminus F$ has an open neighbourhood contained in $X \setminus F$. Hence, $X \setminus F$ is open, which means

$$\bigcup_\alpha F_\alpha \text{ is closed in } X.$$

This proves the lemma.

Now we prove the required result. Let C be an arbitrary closed subset of Y. Consider $f^{-1}(C)$. Since $X = \bigcup_\alpha A_\alpha$, we have

$$\begin{aligned} f^{-1}(C) &= \bigcup_\alpha (f^{-1}(C) \cap A_\alpha) \\ &= \bigcup_\alpha (f|_{A_\alpha})^{-1}(C). \end{aligned}$$

Now, since every restriction $f|_{A_\alpha}$ is continuous and C is closed in Y, $(f|_{A_\alpha})^{-1}(C)$ is closed in A_α. Since A_α is closed in X, it follows that

$$F_\alpha := (f|_{A_\alpha})^{-1}(C)$$

is closed in X. Furthermore, since $F_\alpha \subseteq A_\alpha$ and the family $\{A_\alpha\}$ is locally finite, the family $\{F_\alpha\}$ is also locally finite. Therefore, by our lemma, the union

$$f^{-1}(C) = \bigcup_\alpha F_\alpha$$

is closed in X. Since C was an arbitrary closed subset of Y, the preimage of every closed subset of Y is closed in X. Consequently,

$$\boxed{f\colon X \to Y \text{ is continuous.}}$$ □

Problem 4.4

Let $f\colon X \to A$ and $g\colon Y \to B$ be two continuous maps. Define $f \times g\colon X \times Y \to A \times B$ by $(f \times g)(x, y) = (f(x), g(y))$. Prove that $f \times g$ is a continuous map.

Solution

submitted by **Dhrubojoti**

Proof. We are given that

$$f\colon X \to A \quad \text{and} \quad g\colon Y \to B$$

are continuous maps. We need to prove that

$$f \times g\colon X \times Y \to A \times B$$

is continuous. Recall that the product topology on $A \times B$ has a basis consisting of sets of the form $U \times V$, where U is open in A and V is open in B. We will use the characterization of continuity in terms of the preimages of basic open sets.

Step 1. Consider an arbitrary basic open set in $A \times B$. Let $U \times V$ be an arbitrary basic open set in $A \times B$, where $U \subseteq A$ and $V \subseteq B$ are open. We have

$$\begin{aligned}(f \times g)^{-1}(U \times V) &= \{(x, y) \in X \times Y : (f \times g)(x, y) \in U \times V\} \\ &= \{(x, y) \in X \times Y : (f(x), g(y)) \in U \times V\} \\ &= \{(x, y) \in X \times Y : f(x) \in U,\ g(y) \in V\} \\ &= f^{-1}(U) \times g^{-1}(V).\end{aligned}$$

Therefore,

$$(f \times g)^{-1}(U \times V) = f^{-1}(U) \times g^{-1}(V).$$

Step 2. Show that the preimage is open in $X \times Y$. Since $f\colon X \to A$ is continuous and U is open in A, we know that $f^{-1}(U)$ is open in X. Similarly, since $g\colon Y \to B$ is continuous and V is open in B, $g^{-1}(V)$ is open in Y. Hence, $f^{-1}(U) \times g^{-1}(V)$ is open in $X \times Y$, by the definition of the product topology. Consequently, $(f \times g)^{-1}(U \times V)$ is open in $X \times Y$.

Step 3. Conclude continuity. We have shown that the preimage under $f \times g$ of every basic open set in $A \times B$ is open in $X \times Y$. Since every open subset of $A \times B$ is a union of basic open sets, the preimage of every open subset of $A \times B$ is open in $X \times Y$. Therefore,

$$\boxed{f \times g\colon X \times Y \to A \times B \text{ is continuous.}}$$ □

Problem 4.5

Let (X, d) be a metric space and let $S \subseteq X$ be a non-empty subset. For any $x \in X$, define the distance from x to S by $d(x, S) = \inf_{s \in S} d(x, s)$.

(1) Show that for any $x, y \in X$, $|d(x, S) - d(y, S)| \leq d(x, y)$.

(2) Conclude that $f_S \colon X \to \mathbb{R}$ defined by $f_S(x) = d(x, S)$ is continuous on X.

Solution

submitted by **Aaditya**

Proof.

(1) By the triangle inequality, for every $s \in S$,

$$d(x, s) \leq d(x, y) + d(y, s).$$

Taking the infimum over $s \in S$, we get

$$d(x, S) \leq d(x, y) + d(y, S).$$

Thus,

$$d(x, S) - d(y, S) \leq d(x, y).$$

Interchanging x and y gives

$$d(y, S) - d(x, S) \leq d(x, y).$$

Therefore,

$$\boxed{|d(x, S) - d(y, S)| \leq d(x, y).}$$

(2) Let $x_0 \in X$ and $\varepsilon > 0$. Choose $\delta = \varepsilon$.

If $d(x, x_0) < \delta$, then by (1),

$$\begin{aligned} |f_S(x) - f_S(x_0)| &= |d(x, S) - d(x_0, S)| \\ &\leq d(x, x_0) \\ &< \varepsilon. \end{aligned}$$

Hence, f_S is continuous on X. □

Problem 4.6

Let $\{A_n\}_{n\in\mathbb{N}}$ be a sequence of connected subspaces of X such that $A_n \cap A_{n+1} \neq \emptyset$ for all $n \in \mathbb{N}$. Prove that $\bigcup_{n=1}^{\infty} A_n$ is connected.

Solution

submitted by **Aaditya**

Proof.

Let

$$A = \bigcup_{n=1}^{\infty} A_n.$$

Suppose, for contradiction, that A is disconnected. Then

$$A = U \cup V,$$

where U, V are non-empty, disjoint and open in A.

Since each A_n is connected and

$$A_n = (A_n \cap U) \cup (A_n \cap V),$$

we must have

$$A_n \subseteq U \quad \text{or} \quad A_n \subseteq V.$$

Without loss of generality, let $A_1 \subseteq U$.

Since $A_1 \cap A_2 \neq \emptyset$, there exists $p \in A_1 \cap A_2$. Thus $p \in U$, so $A_2 \not\subseteq V$. Consequently, $A_2 \subseteq U$.

Repeating this argument inductively, we obtain

$$A_n \subseteq U \quad \text{for every } n \in \mathbb{N}.$$

Therefore,

$$A = \bigcup_{n=1}^{\infty} A_n \subseteq U,$$

contradicting $V \neq \emptyset$.

Hence,

$$\boxed{\bigcup_{n=1}^{\infty} A_n \text{ is connected.}}$$

□

Problem 4.7

Determine whether or not $\mathbb{R}^\omega$ is connected under the uniform topology.

Solution

submitted by **Aaditya**

Proof.

The uniform metric is

$$\bar{\rho}(x, y) = \sup_n \min\{|x_n - y_n|, 1\}.$$

Define

$$\mathcal{B} = \left\{ x \in \mathbb{R}^\omega : \sup_n |x_n| < \infty \right\}$$

and

$$\mathcal{U} = \mathbb{R}^\omega \setminus \mathcal{B}.$$

We show that both sets are open.

First, observe that

$$\bar{\rho}(x, y) < 1 \implies |x_n - y_n| < 1 \quad \forall n.$$

Claim 1: $\mathcal{B}$ is open.

Let $x \in \mathcal{B}$. Then $|x_n| \leq M$ for some $M > 0$ and all n.

If $\bar{\rho}(x, y) < 1$, then

$$|y_n| \leq |x_n| + |y_n - x_n| < M + 1.$$

Hence $y \in \mathcal{B}$, proving that $\mathcal{B}$ is open.

Claim 2: $\mathcal{U}$ is open.

Let $x \in \mathcal{U}$ and suppose $\bar{\rho}(x, y) < 1$.

If y were bounded, say $|y_n| \leq M$, then

$$|x_n| \leq |y_n| + |x_n - y_n| < M + 1.$$

This contradicts the fact that x is unbounded.

Thus $y \in \mathcal{U}$, proving that $\mathcal{U}$ is open.

Finally, $\mathcal{B}$ and $\mathcal{U}$ are disjoint, non-empty, and

$$\mathcal{B} \cup \mathcal{U} = \mathbb{R}^\omega.$$

Therefore, they form a separation of $\mathbb{R}^\omega$. Hence,

$$\boxed{\mathbb{R}^\omega \text{ is not connected.}}$$

□

Problem 4.8

Determine whether or not the lower limit topology $\mathbb{R}_\ell$ on $\mathbb{R}$ is connected.

Solution

submitted by **Aaditya**

Proof.

The lower limit topology has the basis

$$\mathcal{B} = \{[a, b) : a < b\}.$$

Consider

$$U = (-\infty, 0), \qquad V = [0, \infty).$$

We show that both sets are open.

If $x \in U$, then

$$x \in [x, 0) \subseteq U.$$

Since $[x, 0)$ is a basis element, U is open.

Similarly, if $x \in V$, then

$$x \in [x, x + 1) \subseteq V.$$

Thus, V is also open.

Moreover,

$$U \cap V = \emptyset, \qquad U \cup V = \mathbb{R}.$$

Both sets are non-empty. Therefore, (U, V) forms a separation of $\mathbb{R}_\ell$.

Hence,

$$\boxed{\mathbb{R}_\ell \text{ is not connected.}}$$

□

Problem 4.9

Let $C \subset X$ be a connected subspace of X and $A \subset X$. Prove that if C intersects both A and $X \setminus A$, then C intersects $\operatorname{Bd} A$.

Solution

submitted by **Harisankar B**

Recall that $\operatorname{Bd} A = \overline{A} \cap \overline{X \setminus A}$, and that $X \setminus \overline{X \setminus A} = \operatorname{Int} A$ and $X \setminus \overline{A} = \operatorname{Int}(X \setminus A)$.

Proof. Suppose, for contradiction, that $C \cap \operatorname{Bd} A = \emptyset$. Consider the two sets

$$P = C \cap \operatorname{Int} A, \qquad Q = C \cap (X \setminus \overline{A}).$$

Both $\operatorname{Int} A$ and $X \setminus \overline{A}$ are open in X, so P and Q are open in the subspace C. They are disjoint, since $\operatorname{Int} A \subseteq A \subseteq \overline{A}$.

$P \cup Q = C$. Let $c \in C$. If $c \in \overline{A}$, then $c \notin \operatorname{Bd} A$ forces $c \notin \overline{X \setminus A}$, i.e. $c \in \operatorname{Int} A$, so $c \in P$. Otherwise $c \in X \setminus \overline{A}$, so $c \in Q$.

$P \neq \emptyset$. Pick $a \in C \cap A$. Then $a \in \overline{A}$, and as above $a \notin \operatorname{Bd} A$ gives $a \in \operatorname{Int} A$; hence $a \in P$.

$Q \neq \emptyset$. Pick $b \in C \cap (X \setminus A)$. Then $b \in \overline{X \setminus A}$, and $b \notin \operatorname{Bd} A$ forces $b \notin \overline{A}$; hence $b \in Q$.

Thus P, Q form a separation of C, contradicting the connectedness of C. Therefore $C \cap \operatorname{Bd} A \neq \emptyset$. □

Problem 4.10

Let X and Y be two connected subspaces and $A \subsetneq X$ and $B \subsetneq Y$. Prove that $(X \times Y) \setminus (A \times B)$ is connected.

Solution

submitted by **Harisankar B**

Let $Z = (X \times Y) \setminus (A \times B)$. We use two standard facts:

(a) For fixed $x \in X$ the map $y \mapsto (x, y)$ is an embedding of Y onto $\{x\} \times Y$ (Problem 4.1); hence $\{x\} \times Y$ is connected. Likewise $X \times \{y\}$ is connected for every $y \in Y$.

(b) A union of connected subspaces having a point in common is connected; in particular, the union of two connected subspaces that intersect is connected.

Proof. Since $A \subsetneq X$ and $B \subsetneq Y$, we can fix $x_0 \in X \setminus A$ and $y_0 \in Y \setminus B$. Note that a point (x, y) lies in $A \times B$ only if $x \in A$ *and* $y \in B$; hence

$$Z = \{(x, y) \in X \times Y : x \notin A \text{ or } y \notin B\}.$$

For $x \in X \setminus A$ and $y \in Y \setminus B$ define

$$T_x = (\{x\} \times Y) \cup (X \times \{y_0\}), \qquad S_y = (X \times \{y\}) \cup (\{x_0\} \times Y).$$

Each T_x and S_y lies in Z. Points of $\{x\} \times Y$ have first coordinate $x \notin A$; points of $X \times \{y_0\}$ have second coordinate $y_0 \notin B$. Similarly for S_y, using $y \notin B$ and $x_0 \notin A$.

Each T_x and S_y is connected. T_x is the union of the connected sets $\{x\} \times Y$ and $X \times \{y_0\}$ (fact (a)), which meet at (x, y_0); so T_x is connected by fact (b). Likewise S_y is the union of the connected sets $X \times \{y\}$ and $\{x_0\} \times Y$, which meet at (x_0, y).

They share a common point. Every T_x contains $X \times \{y_0\} \ni (x_0, y_0)$, and every S_y contains $\{x_0\} \times Y \ni (x_0, y_0)$.

They cover Z. If $(x, y) \in Z$ with $x \notin A$, then $(x, y) \in \{x\} \times Y \subseteq T_x$; if $y \notin B$, then $(x, y) \in X \times \{y\} \subseteq S_y$.

Therefore

$$Z = \bigcup_{x \in X \setminus A} T_x \cup \bigcup_{y \in Y \setminus B} S_y$$

is a union of connected subspaces all containing the point (x_0, y_0), and hence Z is connected by fact (b). □

Problem 4.11

Let $Y \subset X$; and let X and Y be connected. Prove that if A and B form a separation of $X \setminus Y$, then both $Y \cup A$ and $Y \cup B$ are connected.

Solution

submitted by **Harisankar B**

We use the following characterisation of separations (Munkres, Lemma 23.1): *if W is a subspace of X, then a separation of W is a pair of disjoint non-empty sets whose union is W, neither of which contains a limit point (in X) of the other.* We also use that if Y is connected and $W = C \cup D$ is a separation with $Y \subseteq W$, then Y lies entirely in C or entirely in D.

Proof. We show that $Y \cup A$ is connected; the argument for $Y \cup B$ is identical with A and B interchanged.

Suppose, for contradiction, that C, D form a separation of $Y \cup A$. Since Y is connected, Y lies entirely in one of them; say $Y \subseteq C$. Then $D \subseteq (Y \cup A) \setminus Y \subseteq A$. We claim that

$$X = (C \cup B) \cup D$$

is a separation of X.

Union and disjointness. $C \cup D = Y \cup A$ and $A \cup B = X \setminus Y$, so $(C \cup B) \cup D = Y \cup A \cup B = X$. Moreover $D \cap C = \emptyset$, and $D \cap B \subseteq A \cap B = \emptyset$.

Non-emptiness. $D \neq \emptyset$ and $C \neq \emptyset$, as C, D is a separation; hence $C \cup B \neq \emptyset$.

No limit points across. By Lemma 23.1 applied to the separations C, D of $Y \cup A$ and A, B of $X \setminus Y$:

- D contains no limit point of C, and C contains no limit point of D;
- A contains no limit point of B, and B contains no limit point of A.

Since $D \subseteq A$, the set D contains no limit point of B; so D contains no limit point of $C \cup B$ (a limit point of a union is a limit point of one of the sets). Also every limit point of D is a limit point of A, so B contains no limit point of D; together with the first bullet, $C \cup B$ contains no limit point of D.

By Lemma 23.1, $C \cup B$ and D form a separation of X, contradicting the connectedness of X. Hence $Y \cup A$ is connected, and similarly $Y \cup B$ is connected. □

Problem Set 5

Problem 5.1

Prove that the punctured Euclidean space $X = \mathbb{R}^n \setminus \{0\}$ is path connected if and only if $n > 1$.

Solution pending

Problem 5.2

Prove that $\mathbb{R}^2 \setminus \mathbb{Q}^2$ with the subspace topology of $\mathbb{R}^2$ is connected, but $\mathbb{R} \setminus \mathbb{Q}$ with the subspace topology of $\mathbb{R}$ is not connected.

Solution pending

Problem 5.3

Let X and Y be topological spaces, and suppose there exist embeddings $f\colon X \to Y$ and $g\colon Y \to X$. Is it necessary that X and Y are homeomorphic? Prove or provide a counterexample.

Solution pending

Problem 5.4

Let X be a non-empty topological space and let $\mathscr{B}$ be a base for the topology on X. Prove that X is compact if and only if every open cover of X consisting of elements of $\mathscr{B}$ has a finite subcover.

Solution pending

Problem 5.5

Let $\mathcal{T}_1$ and $\mathcal{T}_2$ be two topologies on X such that both $(X, \mathcal{T}_1)$ and $(X, \mathcal{T}_2)$ are compact Hausdorff spaces. Prove that either $\mathcal{T}_1 = \mathcal{T}_2$ or they are not comparable.

Solution

submitted by **Prabhu**

Proof. Suppose $\mathcal{T}_1$ and $\mathcal{T}_2$ are comparable. Without loss of generality, let

$$\mathcal{T}_1 \subseteq \mathcal{T}_2.$$

Consider the identity map

$$I \colon (X, \mathcal{T}_2) \to (X, \mathcal{T}_1), \qquad I(x) := x.$$

Clearly, I is *bijective*.

For all $U \in \mathcal{T}_1$, we have

$$I^{-1}(U) = U \in \mathcal{T}_2.$$

Hence, I is *continuous*.

Since I is a bijective continuous map from a compact space to a Hausdorff space, I is a *homeomorphism*. So, I^{-1} is also continuous.

Thus, for all $U \in \mathcal{T}_2$, we have

$$(I^{-1})^{-1}(U) = I(U) = U \in \mathcal{T}_1.$$

Hence,

$$\mathcal{T}_2 \subseteq \mathcal{T}_1.$$

Therefore,

$$\boxed{\mathcal{T}_1 = \mathcal{T}_2.}$$

Hence, either $\mathcal{T}_1 = \mathcal{T}_2$ or they are not comparable. □

Problem 5.6

Prove that every compact subspace of a metric space is closed and bounded in that metric topology. Find an example of a metric space in which a closed and bounded subspace is not compact.

Solution

submitted by **Prabhu**

Proof. Let (X, d) be a metric space and let $K \subseteq X$ be compact.

We know that every metric space is Hausdorff and a compact subspace of a Hausdorff space is closed. Since X is a metric space it is Hausdorff. And K is a compact subspace of X, thus it is *closed* in X.

Let $a \in X$. Consider the collection

$$\mathcal{U} := \{B(a, n) : n \in \mathbb{N}\},$$

where

$$B(a,n) := \{x \in X : d(x,a) < n\}.$$

For all $x \in K$, $d(x,a)$ is finite; so, $\exists\, n \in \mathbb{N}$ such that $d(x,a) < n$; i.e., $x \in B(a,n)$. Thus, $\mathcal{U}$ is an open cover of A.

Since K is compact, there exists a finite subcover $\{B(a,n_1),\ldots,B(a,n_m)\}$. Therefore,

$$K \subseteq \bigcup_{i=1}^{m} B(a,n_i).$$

Let $N := \max\{n_1,n_2,\ldots,n_m\}$. Since $B(a,n_i) \subseteq B(a,N)$ for every i, we get

$$K \subseteq \bigcup_{i=1}^{m} B(a,n_i) \subseteq B(a,N).$$

Therefore, K is *bounded*. Hence,

$$\boxed{K \text{ is closed and bounded.}}$$

□

Example: Consider the metric space $X := (0,1)$ with the usual metric ($\because d(x,y) := |x-y|$), and let $A := X$. Clearly, A is closed ($\because$ it is the entire space) and bounded ($\because \forall x \in A,\ 0 < x < 1$) in X.

However, consider the collection

$$\mathcal{U} = \left\{ \left(\frac{1}{n}, 1\right) : n \in \mathbb{N},\ n \geq 2 \right\}.$$

Each member of $\mathcal{U}$ is open in the subspace $(0,1)$. And,

$$(0,1) = \bigcup_{n=2}^{\infty} \left(\frac{1}{n}, 1\right).$$

Thus, $\mathcal{U}$ is an open cover of A.

For every finite subcollection $\left\{\left(\frac{1}{n_1},1\right), \left(\frac{1}{n_2},1\right), \ldots, \left(\frac{1}{n_k},1\right)\right\}$, let $N = \max\{n_1,n_2,\ldots,n_k\}$. Since $n_i \leq N$ for every $i = 1,\ldots,k$, we have

$$\left(\frac{1}{n_i},1\right) \subseteq \left(\frac{1}{N},1\right).$$

Therefore,

$$\bigcup_{i=1}^{k} \left(\frac{1}{n_i},1\right) \subseteq \left(\frac{1}{N},1\right) \subsetneq (0,1).$$

It cannot cover $(0,1)$, since points in $(0,1/N]$ remain uncovered. Hence, $\mathcal{U}$ has no finite subcover. Therefore,

$$\boxed{(0,1) \text{ is closed and bounded but not compact in } (0,1).}$$

Problem 5.7

Let X be a compact metric space. Prove that if $f\colon X \to X$ is an isometry (distance preserving map) of X, then f is a homeomorphism.

Solution pending

Problem 5.8

Let $\mathbb{R}$ be given the cocountable topology. Is the closed interval $[0, 1]$ a compact subspace of $\mathbb{R}$ under this topology?

Solution pending

Problem 5.9

Let $f\colon X \to Y$ be a continuous map between topological spaces. Show that if X is compact and Y is Hausdorff, then f is a closed map.

Solution pending

Problem 5.10

Prove that if Y is compact, then the projection $\pi_X\colon X \times Y \to X$ is a closed map.

Solution pending

Problem Set 6

Problem 6.1

Let $p\colon X \twoheadrightarrow Y$ be a closed continuous surjective map such that $p^{-1}(y)$ is compact for each $y \in Y$. Prove that if Y is compact, then so is X.

Solution *submitted by* **Yashpal Prasad**

Proof. Let $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ be an arbitrary open covering of the space X. We wish to extract a finite subcovering of X from $\mathcal{U}$ to establish that X is compact. For each point $y \in Y$, the fiber $p^{-1}(y) \subseteq X$ is a compact subspace by hypothesis. Since $\mathcal{U}$ covers X, it covers $p^{-1}(y)$ in particular; hence, we can extract a finite subcollection of open sets from $\mathcal{U}$, say $U_{\alpha_{1,y}}, U_{\alpha_{2,y}}, \ldots, U_{\alpha_{k_y,y}}$, which covers $p^{-1}(y)$.

Define the union of these selected open sets as $S_y = \bigcup_{i=1}^{k_y} U_{\alpha_{i,y}}$, so that $p^{-1}(y) \subseteq S_y$. The complement $V_y = X \setminus S_y$ is a closed subset of X because S_y is a finite union of open sets. Because $p\colon X \to Y$ is a closed map, the image $p(V_y)$ is a closed subset of Y, which implies that its complement $W_y = Y \setminus p(V_y)$ is an open subset of Y.

We claim that $y \in W_y$. Suppose to the contrary that $y \notin W_y$. Then $y \in p(V_y)$, meaning there exists some $x \in V_y$ such that $p(x) = y$, or $x \in p^{-1}(y)$. But $V_y = X \setminus S_y$ is disjoint from S_y, whereas $p^{-1}(y) \subseteq S_y$, yielding a contradiction. Thus, $y \in W_y$ for every $y \in Y$.

The collection of open sets $\{W_y\}_{y \in Y}$ forms an open covering of Y. Since Y is compact, this covering contains a finite subcovering, say $W_{y_1}, W_{y_2}, \ldots, W_{y_m}$. For each j, we have $p^{-1}(W_{y_j}) \subseteq X \setminus V_{y_j} = S_{y_j} = \bigcup_{i=1}^{k_{y_j}} U_{\alpha_{i,y_j}}$. Taking the inverse images over the finite subcovering yields:

$$X = p^{-1}(Y) = \bigcup_{j=1}^{m} p^{-1}(W_{y_j}) \subseteq \bigcup_{j=1}^{m} \left(\bigcup_{i=1}^{k_{y_j}} U_{\alpha_{i,y_j}} \right).$$

Conclusion: We have expressed X as a finite union of sets belonging to $\mathcal{U}$. Therefore, X is compact. □

Problem 6.2

Let X be a compact metric space. Prove that if $f\colon X \to X$ is an isometry (distance preserving map) of X, then f is a homeomorphism.

Solution

submitted by **Yashpal Prasad**

Proof. Let $f\colon X \to X$ satisfy $d(f(x), f(y)) = d(x, y)$ for all $x, y \in X$. A homeomorphism is a continuous bijection with continuous inverse. We check each part in turn.

Step 1: f is continuous. Given $\epsilon > 0$, take $\delta = \epsilon$. If $d(x, y) < \delta$, then $d(f(x), f(y)) = d(x, y) < \epsilon$. So f is (uniformly) continuous.

Step 2: f is injective. If $f(x) = f(y)$, then $d(x, y) = d(f(x), f(y)) = 0$. Hence $x = y$.

Step 3: f is surjective (this is where compactness is used). Suppose, for contradiction, that $f(X) \neq X$, and pick $y \in X \setminus f(X)$.

(a) *$f(X)$ is closed.* By Step 1, f is continuous, and X is compact. The continuous image of a compact set is compact (standard theorem), so $f(X)$ is compact. A compact subset of a metric space is closed, so $f(X)$ is closed.

(b) *y is at positive distance from $f(X)$.* Since $y \notin f(X)$ and $f(X)$ is closed, some open ball $B(y, \epsilon)$ misses $f(X)$. That is, $d(y, z) \geq \epsilon$ for every $z \in f(X)$.

(c) *Build a sequence with no convergent subsequence.* Define $x_n = f^n(y)$ for $n \geq 0$, where $f^0 = \mathrm{id}$. Take $m > n \geq 0$. Since f is an isometry, so is f^n, and therefore

$$d(x_n, x_m) = d\big(f^n(y), f^n(f^{m-n}(y))\big) = d\big(y, f^{m-n}(y)\big).$$

Because $m - n \geq 1$, the point $f^{m-n}(y) = f(f^{m-n-1}(y))$ lies in $f(X)$. By (b), $d(x_n, x_m) \geq \epsilon$.

(d) *Contradiction.* Every pair of distinct terms of (x_n) is at distance at least ϵ. So no subsequence is Cauchy, and hence none converges. But X is a compact metric space, so it is sequentially compact (standard theorem), and (x_n) must have a convergent subsequence. This is a contradiction.

Therefore $f(X) = X$, so f is surjective.

Step 4: f^{-1} is continuous. By Steps 2 and 3, f is a bijection, so $f^{-1}\colon X \to X$ exists. For $a, b \in X$, write $a = f(x)$ and $b = f(y)$. Then

$$d\big(f^{-1}(a), f^{-1}(b)\big) = d(x, y) = d(f(x), f(y)) = d(a, b).$$

So f^{-1} is also an isometry, and by the same argument as in Step 1 it is continuous.

Conclusion: f is a continuous bijection with continuous inverse, so f is a homeomorphism. □

Problem 6.3

Prove that the composition of two quotient maps is a quotient map.

Solution

submitted by **Yashpal Prasad**

Proof. Let $p\colon X \to Y$ and $q\colon Y \to Z$ be quotient maps. By definition, both p and q are continuous and surjective maps. Consider the composite map $qp\colon X \to Z$ defined by $(qp)(x) = q(p(x))$. The composition of continuous functions is continuous, and the composition of surjective functions is surjective; hence qp is continuous and surjective.

To prove that qp is a quotient map, we must show that a subset $U \subseteq Z$ is open in Z if and only if its inverse image $(qp)^{-1}(U)$ is open in X. By set-theoretic properties of inverse images under compositions, $(qp)^{-1}(U) = p^{-1}(q^{-1}(U))$.

Suppose $U \subseteq Z$ is such that $(qp)^{-1}(U)$ is open in X. Since p is a quotient map, a subset $V \subseteq Y$ is open in Y if and only if $p^{-1}(V)$ is open in X. Setting $V = q^{-1}(U)$, we have $p^{-1}(V) = (qp)^{-1}(U)$ open in X, which implies $q^{-1}(U)$ is open in Y. Now, since q is a quotient map and $q^{-1}(U)$ is open in Y, it follows that U is open in Z. Conversely, if U is open in Z, then $q^{-1}(U)$ is open in Y by continuity of q, and $p^{-1}(q^{-1}(U))$ is open in X by continuity of p.

Conclusion: U is open in Z if and only if $(qp)^{-1}(U)$ is open in X. Thus, qp is a quotient map. □

Problem 6.4

Prove that a surjective continuous map $f\colon X \twoheadrightarrow Y$ that is either an open map or a closed map is a quotient map.

Solution

submitted by **Indrajeet Kumawat**

Proof. A surjective continuous map $f\colon X \to Y$ is a quotient map if for any subset $U \subseteq Y$, U is open in Y if and only if $f^{-1}(U)$ is open in X (equivalently with closed sets). We analyze the two cases separately:

Case 1 (f is an open map): Let $U \subseteq Y$ be any subset. If U is open in Y, then $f^{-1}(U)$ is open in X by continuity. Conversely, suppose $f^{-1}(U)$ is open in X. Since f is an open map, the image $f(f^{-1}(U))$ is open in Y. Because f is surjective, $f(f^{-1}(U)) = U$, so U is open in Y.

Case 2 (f is a closed map): Let $C \subseteq Y$ be any subset. If C is closed in Y, then $f^{-1}(C)$ is closed in X by continuity. Conversely, suppose $f^{-1}(C)$ is closed in X. Since f is a closed map, $f(f^{-1}(C))$ is closed in Y. Since f is surjective, $f(f^{-1}(C)) = C$, so C is closed in Y.

Conclusion: In both cases, the quotient map condition holds. Therefore, f is a quotient map. □

Problem 6.5

Prove that a bijective quotient map is a homeomorphism.

Solution *submitted by* **Indrajeet Kumawat**

Proof. Let $f\colon X \to Y$ be a quotient map that is also bijective. Because f is a quotient map, it is continuous and surjective. Because f is bijective, it possesses an inverse function $f^{-1}\colon Y \to X$. To prove that f is a homeomorphism, we must show that f^{-1} is continuous (since f is already continuous).

Showing f^{-1} is continuous is equivalent to showing that for any open set $U \subseteq X$, its inverse image under f^{-1}, which is $(f^{-1})^{-1}(U) = f(U)$, is open in Y. Let U be an arbitrary open subset of X. Because f is injective, $f^{-1}(f(U)) = U$. By definition, f is a quotient map, meaning a subset $V \subseteq Y$ is open in Y if and only if $f^{-1}(V)$ is open in X. Choosing $V = f(U)$, we have $f^{-1}(V) = U$, which is given to be open in X. Therefore, $V = f(U)$ is open in Y.

Conclusion: Since $f(U)$ is open in Y for every open set $U \subseteq X$, the inverse map f^{-1} is continuous. Thus, f is a homeomorphism. □

Problem 6.6

Let $f_1\colon X_1 \to Y_1$ and $f_2\colon X_2 \to Y_2$ be maps between topological spaces, and let $f_1 \times f_2\colon X_1 \times X_2 \to Y_1 \times Y_2$ denote the product map $(x_1, x_2) \mapsto (f_1(x_1), f_2(x_2))$.

(a) Prove that if f_1 and f_2 are continuous, then $f_1 \times f_2$ is continuous with respect to the product topologies.

(b) Give a counterexample to show that if $p_1\colon X_1 \to Y_1$ and $p_2\colon X_2 \to Y_2$ are quotient maps, $p_1 \times p_2$ need not be a quotient map.

Solution *submitted by* **Indrajeet Kumawat**

(a) A sub-basis for the product topology on $Y_1 \times Y_2$ consists of sets of the form $U_1 \times U_2$, where $U_1 \subseteq Y_1$ and $U_2 \subseteq Y_2$ are open. To prove $f_1 \times f_2$ is continuous, we show that the inverse image of any basic open set $U_1 \times U_2$ is open in $X_1 \times X_2$. Let $(x_1, x_2) \in (f_1 \times f_2)^{-1}(U_1 \times U_2)$. Then $(f_1(x_1), f_2(x_2)) \in U_1 \times U_2$, which implies $f_1(x_1) \in U_1$ and $f_2(x_2) \in U_2$, or equivalently $x_1 \in f_1^{-1}(U_1)$ and $x_2 \in f_2^{-1}(U_2)$. Thus, $(f_1 \times f_2)^{-1}(U_1 \times U_2) = f_1^{-1}(U_1) \times f_2^{-1}(U_2)$. Since f_1 and f_2 are continuous, $f_1^{-1}(U_1)$ and $f_2^{-1}(U_2)$ are open in X_1 and X_2 respectively, making their Cartesian product open in $X_1 \times X_2$.

Conclusion (a): $f_1 \times f_2$ is continuous. □

(b) Consider the quotient map $p_1 = \pi\colon \mathbb{R} \to \mathbb{R}/\mathbb{N}$ obtained by collapsing the closed set of integers $\mathbb{N} = \{1, 2, 3, \dots\}$ to a single point $*$, and let $p_2 = \mathrm{id}_{\mathbb{Q}}\colon \mathbb{Q} \to \mathbb{Q}$ be the identity map on the rational numbers, which is a homeomorphism and hence a quotient map. We examine their product map $F = \pi \times \mathrm{id}_{\mathbb{Q}}\colon \mathbb{R} \times \mathbb{Q} \to (\mathbb{R}/\mathbb{N}) \times \mathbb{Q}$. By part (a), F is continuous, and it is clearly surjective. However, we claim that F is not a quotient map.

To see why, note that the fibers of F over points in the codomain are either single points or sets of the form $\mathbb{N} \times \{s\}$ for $s \in \mathbb{Q}$. A subset $U \subseteq \mathbb{R} \times \mathbb{Q}$ is saturated with respect to F if for each rational s, either the entire vertical slice $\mathbb{N} \times \{s\}$ is contained in U or none of it is. We can construct a saturated open set U by taking an enumeration of irrational numbers $r_n = \sqrt{2}/n$ for $n \geq 1$ and defining continuous functions $h_n(s) = \min\{1/4, |s - r_n|\}$ for $s \in \mathbb{Q}$. Because each r_n is irrational, $h_n(s) > 0$ everywhere on $\mathbb{Q}$. We define the open set

$$U = \bigcup_{n=1}^{\infty} \{(x, s) \in \mathbb{R} \times \mathbb{Q} : |x - n| < h_n(s)\}.$$

Since $(n, s) \in U$ for every integer n and rational s, the set contains $\mathbb{N} \times \mathbb{Q}$, ensuring it is saturated.

Let $W = F(U)$ be the image of U. Because U is saturated, the inverse image $F^{-1}(W) = U$ is open in $\mathbb{R} \times \mathbb{Q}$. If F were a quotient map, its image W would necessarily be open in $(\mathbb{R}/\mathbb{N}) \times \mathbb{Q}$. Notice that the point $(*, 0)$ belongs to W. If W were open, there would exist an open neighborhood V of $*$ in $\mathbb{R}/\mathbb{N}$ and a positive number $\delta > 0$ such that $V \times ((-\delta, \delta) \cap \mathbb{Q}) \subseteq W$. The inverse image $\pi^{-1}(V)$ is an open neighborhood in $\mathbb{R}$ containing all integers $n \geq 1$, meaning for each n there exists some $0 < \eta_n < 1/4$ such that $(n - \eta_n, n + \eta_n) \subseteq \pi^{-1}(V)$. By choosing n sufficiently large such that $|r_n| < \delta/2$ and picking a rational s close to r_n via the density of $\mathbb{Q}$, we can construct a point $x = n + (|s - r_n| + \eta_n)/2$ that satisfies $|s - r_n| < |x - n| < \eta_n$ while simultaneously contradicting the bound established by $h_n(s)$. This contradiction shows that the image W cannot be open in the product topology.

Conclusion (b): The product of quotient maps $p_1 \times p_2$ is not necessarily a quotient map. □

Problem 6.7

Let X and Y be topological spaces, and $f\colon X \to Y$ be a bijective continuous map. Show that if Y is Hausdorff, then X is also Hausdorff.

Solution

submitted by **Himanshu Sharma**

Proof. To show that X is Hausdorff, we must demonstrate that for any two distinct points $x_1, x_2 \in X$ with $x_1 \neq x_2$, there exist disjoint open neighborhoods $U_1, U_2 \subseteq X$ containing x_1 and x_2 respectively. Since $f\colon X \to Y$ is injective, $f(x_1) \neq f(x_2)$. Because Y is a Hausdorff space, there exist disjoint open sets $V_1, V_2 \subseteq Y$ such that $f(x_1) \in V_1$, $f(x_2) \in V_2$, and $V_1 \cap V_2 = \emptyset$.

Since f is continuous, the inverse images $U_1 = f^{-1}(V_1)$ and $U_2 = f^{-1}(V_2)$ are open subsets of X containing x_1 and x_2 respectively. Furthermore, their intersection is:

$$U_1 \cap U_2 = f^{-1}(V_1) \cap f^{-1}(V_2) = f^{-1}(V_1 \cap V_2) = f^{-1}(\emptyset) = \emptyset.$$

Conclusion: We have found disjoint open neighborhoods separating x_1 and x_2. Therefore, X is Hausdorff. □

Problem 6.8

Let X be a compact topological space and $p\colon X \twoheadrightarrow Y$ be a quotient map.

(a) Prove that Y is compact.

(b) Show that if Y is Hausdorff, then p is a closed map.

Solution *submitted by* **Himanshu Sharma**

(a) A quotient map $p\colon X \to Y$ is continuous and surjective. It is a fundamental topological theorem that the continuous image of a compact space is compact. Since X is compact and $Y = p(X)$, Y is compact.

Conclusion (a): Y is compact. □

(b) Let $C \subseteq X$ be an arbitrary closed set in X. We must prove that $p(C)$ is closed in Y. Because X is compact and C is closed in X, C is compact. Consider the restriction map $p|_C\colon C \to p(C)$, which is continuous. Thus, $p(C) = p|_C(C)$ is the continuous image of a compact space, so $p(C)$ is compact in Y. Since Y is Hausdorff, any compact subspace of a Hausdorff space is closed. Consequently, $p(C)$ is closed in Y.

Conclusion (b): p is a closed map. □

Problem 6.9

Let $p\colon X \to Y$ be a continuous surjection. Prove that p is a closed map if and only if for every subset $S \subseteq Y$ and every open set $U \subseteq X$ containing $p^{-1}(S)$, there exists an open set $V \subseteq Y$ containing S such that $p^{-1}(V) \subseteq U$.

Solution *submitted by* **Himanshu Sharma**

Proof. **Proof of (⇒):** Suppose p is a closed map. Let $S \subseteq Y$ and let $U \subseteq X$ be an open set containing $p^{-1}(S)$. Consider the closed set $C = X \setminus U$. Since p is a closed map, $p(C)$ is closed in Y, so $V = Y \setminus p(C)$ is open in Y. For any $s \in S$, $p^{-1}(\{s\}) \subseteq p^{-1}(S) \subseteq U$, meaning no point in the fiber belongs to C. Thus $s \notin p(C)$, so $s \in V$, implying $S \subseteq V$. Furthermore, $p^{-1}(V) = X \setminus p^{-1}(p(C)) \subseteq X \setminus C = U$.

Proof of (⇐): Conversely, suppose the condition holds. Let $C \subseteq X$ be closed. We must show $p(C)$ is closed in Y, or $Y \setminus p(C)$ is open. Let $y \in Y \setminus p(C)$ and set $S = \{y\}$. The fiber $p^{-1}(\{y\})$ is disjoint from C, so $p^{-1}(\{y\}) \subseteq X \setminus C$. Let $U = X \setminus C$, which is open. By hypothesis, there exists an open set $V_y \subseteq Y$ containing y such that $p^{-1}(V_y) \subseteq U = X \setminus C$. This implies $V_y \cap p(C) = \emptyset$, so $V_y \subseteq Y \setminus p(C)$. Thus, $Y \setminus p(C)$ is open.

Conclusion: p is a closed map if and only if the stated condition holds. □

Problem Set 7 (Munkres §22)

Problem 7.1

Check the details of Example 3.[a]

[a] Example 3 of §22 (The Quotient Topology) in J. R. Munkres, *Topology*, 2nd ed.

Solution *submitted by* **Karan Gangadharan**

The quotient topology on A induced by p is given by

$$\tau_A = \{U \subseteq A : p^{-1}(U) \text{ is open in } \mathbb{R}\}.$$

Since

$$p^{-1}(\{a\}) = (-\infty, 0), \qquad p^{-1}(\{b\}) = (0, \infty), \qquad p^{-1}(\{c\}) = \{0\},$$

and

$$p^{-1}(\{a, b\}) = \mathbb{R} \setminus \{0\},$$

we observe that the open subsets of A are

$$\varnothing, \quad \{a\}, \quad \{b\}, \quad \{a, b\}, \quad \{a, b, c\}.$$

Hence, the quotient topology on A is

$$\boxed{\tau_A = \{\varnothing, \{a\}, \{b\}, \{a, b\}, \{a, b, c\}\}.}$$

Problem 7.2

(a) Let $p\colon X \to Y$ be a continuous map. Show that if there is a continuous map $f\colon Y \to X$ such that $p \circ f$ equals the identity map of Y, then p is a quotient map.

(b) If $A \subset X$, a *retraction* of X onto A is a continuous map $r\colon X \to A$ such that $r(a) = a$ for each $a \in A$. Show that a retraction is a quotient map.

Solution *submitted by* **Karan Gangadharan**

Proof. **(a)** Let $p\colon X \to Y$ be a continuous map. Suppose there exists a continuous map $f\colon Y \to X$ such that

$$p \circ f = I_Y,$$

where I_Y is the identity map on Y.

We know that p is continuous. Also, since

$$p(f(y)) = y, \qquad \forall y \in Y,$$

the map p is surjective.

Now, let $U \subseteq Y$ be such that $p^{-1}(U)$ is open in X. Since f is continuous, we have $f^{-1}\big(p^{-1}(U)\big)$ open in Y. But

$$f^{-1}\big(p^{-1}(U)\big) = (p \circ f)^{-1}(U) = I_Y^{-1}(U) = U.$$

Therefore, U is open in Y.

Thus, p is a continuous surjective map satisfying

$$U \text{ is open in } Y \iff p^{-1}(U) \text{ is open in } X.$$

Hence, p is a quotient map.

$\boxed{p \text{ is a quotient map.}}$

(b) Let $A \subseteq X$ and let $r\colon X \to A$ be a retraction such that

$$r(a) = a, \qquad \forall a \in A.$$

We know that r is continuous by definition. Also, since $r(a) = a$ for every $a \in A$, the map r is surjective.

Now, let $U \subseteq A$ be such that $r^{-1}(U)$ is open in X. Since A has the subspace topology, we have $r^{-1}(U) \cap A$ open in A. But, using the property of retraction,

$$\begin{aligned} r^{-1}(U) \cap A &= \{x \in A : r(x) \in U\} \\ &= \{x \in A : x \in U\} \\ &= U. \end{aligned}$$

Therefore, U is open in A.

Thus, r is a continuous surjective map such that

$$U \text{ is open in } A \iff r^{-1}(U) \text{ is open in } X.$$

Hence, r is a quotient map.

$\boxed{\text{Every retraction is a quotient map.}}$ □

Problem 7.3

Let $\pi_1\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be projection on the first coordinate. Let A be the subspace of $\mathbb{R} \times \mathbb{R}$ consisting of all points $x \times y$ for which either $x \geq 0$ or $y = 0$ (or both); let $q\colon A \to \mathbb{R}$ be obtained by restricting π_1. Show that q is a quotient map that is neither open nor closed.

Solution

submitted by **Karan Gangadharan**

Proof. Let

$$\pi_1\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$$

be the projection onto the first coordinate, defined by $\pi_1(x, y) = x$. Let

$$A = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x \geq 0 \text{ or } y = 0\}.$$

Define $g\colon A \to \mathbb{R}$ by restricting π_1 to A. Thus, $g(x, y) = x$.

We shall show that g is a quotient map but is neither open nor closed.

(i) g is a quotient map: Since g is the restriction of the continuous projection map π_1, it is continuous. Also, g is surjective, because for every $x \in \mathbb{R}$,

$$(x, 0) \in A \quad \text{and} \quad g(x, 0) = x.$$

Now, consider the map $f\colon \mathbb{R} \to A$ defined by $f(x) = (x, 0)$. Clearly, f is continuous and

$$(g \circ f)(x) = g(x, 0) = x.$$

Therefore, $g \circ f = I_{\mathbb{R}}$. By Problem 7.2(a), since g and f are continuous and $g \circ f = I_{\mathbb{R}}$, it follows that

$$\boxed{g \text{ is a quotient map.}}$$

(ii) g is not an open map: Consider the set

$$U = A \cap ((-1, 1) \times (1/2, 3/2)).$$

Since $(-1, 1) \times (1/2, 3/2)$ is open in $\mathbb{R}^2$, U is open in A. Now, by the definition of A, if $(x, y) \in U$, then $x \geq 0$, since $y \neq 0$. Consequently,

$$g(U) = [0, 1).$$

But $[0, 1)$ is not open in $\mathbb{R}$. Thus, g does not map every open subset of A onto an open subset of $\mathbb{R}$. Hence,

$$\boxed{g \text{ is not an open map.}}$$

(iii) g is not a closed map: Consider the subset

$$F = \left\{ \left(\frac{1}{n}, n \right) : n \in \mathbb{N} \right\}.$$

Clearly, $F \subseteq A$, since $1/n > 0$ for every $n \in \mathbb{N}$. We claim that F is closed in A. Indeed, F is closed in $\mathbb{R}^2$, because the second coordinates of its points tend to infinity, so it has no accumulation point in $\mathbb{R}^2$. Therefore, F is also closed in the subspace A. Now,

$$g(F) = g\left(\left\{ \left(\frac{1}{n}, n \right) : n \in \mathbb{N} \right\} \right) = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}.$$

But

$$\frac{1}{n} \longrightarrow 0 \quad \text{as } n \longrightarrow \infty,$$

and $0 \notin g(F)$. Hence, $g(F)$ is not closed in $\mathbb{R}$. Therefore, g does not map every closed subset of A onto a closed subset of $\mathbb{R}$. Thus,

$$\boxed{g \text{ is not a closed map.}}$$

Combining (i), (ii) and (iii), we conclude that

$$\boxed{g\colon A \to \mathbb{R} \text{ is a quotient map which is neither open nor closed.}} \qquad \square$$

Problem 7.4

(a) Define an equivalence relation on the plane $X = \mathbb{R}^2$ as follows:

$$x_0 \times y_0 \sim x_1 \times y_1 \quad \text{if} \quad x_0 + y_0^2 = x_1 + y_1^2.$$

Let X^* be the corresponding quotient space. It is homeomorphic to a familiar space; what is it? [*Hint:* Set $g(x \times y) = x + y^2$.]

(b) Repeat (a) for the equivalence relation

$$x_0 \times y_0 \sim x_1 \times y_1 \quad \text{if} \quad x_0^2 + y_0^2 = x_1^2 + y_1^2.$$

Solution *submitted by* **Karan Gangadharan**

Proof. **(a)** Define an equivalence relation on $X = \mathbb{R}^2$ by

$$(x_0, y_0) \sim (x_1, y_1) \iff x_0 + y_0^2 = x_1 + y_1^2.$$

Let $X^* = X/\sim$ be the corresponding quotient space. We shall show that X^* is homeomorphic to $\mathbb{R}$.

Define a map $g\colon X \to \mathbb{R}$ by

$$g(x, y) = x + y^2.$$

Since g is a polynomial function, it is continuous. Also, g is surjective, since for every $t \in \mathbb{R}$, $g(t, 0) = t$.

Now, define a map $f\colon \mathbb{R} \to X$ by $f(t) = (t, 0)$. Clearly, f is continuous and

$$(g \circ f)(t) = g(t, 0) = t.$$

Thus, $g \circ f = I_{\mathbb{R}}$. Therefore, by Problem 7.2(a), g is a quotient map.

Let $q\colon X \to X^*$ be the natural quotient map defined by $q(x, y) = [(x, y)]$. Define $\overline{g}\colon X^* \to \mathbb{R}$ by

$$\overline{g}([(x, y)]) = x + y^2.$$

We show that $\overline{g}$ is a homeomorphism.

(i) $\overline{g}$ is well-defined: Suppose $[(x_0, y_0)] = [(x_1, y_1)]$. Then, by the definition of the equivalence relation, $x_0 + y_0^2 = x_1 + y_1^2$. Hence, $\overline{g}([(x_0, y_0)]) = \overline{g}([(x_1, y_1)])$. Therefore, $\overline{g}$ is well-defined.

(ii) $\overline{g}$ is bijective: Suppose $\overline{g}([(x_0, y_0)]) = \overline{g}([(x_1, y_1)])$. Then $x_0 + y_0^2 = x_1 + y_1^2$. Thus, $(x_0, y_0) \sim (x_1, y_1)$, which implies $[(x_0, y_0)] = [(x_1, y_1)]$. Therefore, $\overline{g}$ is injective. Further, since g is surjective, for every $t \in \mathbb{R}$, there exists $(x, y) \in X$ such that $\overline{g}([(x, y)]) = t$. Hence, $\overline{g}$ is surjective. Therefore, $\overline{g}$ is bijective.

(iii) $\overline{g}$ is a homeomorphism: We have

$$g = \overline{g} \circ q.$$

Since g is continuous and q is a quotient map, the quotient property of q implies that $\overline{g}$ is continuous.

Also, let U be any open subset of X^*. Since $\overline{g}$ is injective, we have

$$g^{-1}(\overline{g}(U)) = q^{-1}(U).$$

Since q is a quotient map and U is open in X^*, $q^{-1}(U)$ is open in X. Thus, $g^{-1}(\overline{g}(U))$ is open in X. Since g is a quotient map, $\overline{g}(U)$ is open in $\mathbb{R}$. Therefore, $\overline{g}$ is an open map.

Hence, $\overline{g}$ is a continuous bijective open map, and consequently it is a homeomorphism. Thus,

$$\boxed{X^* \cong \mathbb{R}.}$$

(b) Define an equivalence relation on $X = \mathbb{R}^2$ by

$$(x_0, y_0) \sim (x_1, y_1) \iff x_0^2 + y_0^2 = x_1^2 + y_1^2.$$

Let $X^* = X/\sim$ be the corresponding quotient space. We shall show that X^* is homeomorphic to $[0, \infty)$.

Define a map $g\colon X \to [0, \infty)$ by

$$g(x, y) = x^2 + y^2.$$

Since g is a polynomial function, it is continuous. Also, g is surjective, since for every $t \in [0, \infty)$, $g(\sqrt{t}, 0) = t$.

Now, define a map $f\colon [0, \infty) \to X$ by $f(t) = (\sqrt{t}, 0)$. Clearly, f is continuous and

$$(g \circ f)(t) = g(\sqrt{t}, 0) = t.$$

Thus, $g \circ f = I_{[0,\infty)}$. Therefore, by Problem 7.2(a), g is a quotient map.

Let $q\colon X \to X^*$ be the natural quotient map defined by $q(x, y) = [(x, y)]$. Define $\overline{g}\colon X^* \to [0, \infty)$ by

$$\overline{g}([(x, y)]) = x^2 + y^2.$$

We show that $\overline{g}$ is a homeomorphism.

(i) $\overline{g}$ is well-defined: Suppose $[(x_0, y_0)] = [(x_1, y_1)]$. Then, by the definition of the equivalence relation, $x_0^2 + y_0^2 = x_1^2 + y_1^2$. Hence, $\overline{g}([(x_0, y_0)]) = \overline{g}([(x_1, y_1)])$. Therefore, $\overline{g}$ is well-defined.

(ii) $\overline{g}$ is bijective: Suppose $\overline{g}([(x_0, y_0)]) = \overline{g}([(x_1, y_1)])$. Then $x_0^2 + y_0^2 = x_1^2 + y_1^2$. Thus, $(x_0, y_0) \sim (x_1, y_1)$, which implies $[(x_0, y_0)] = [(x_1, y_1)]$. Therefore, $\overline{g}$ is injective. Further, since g is surjective, for every $t \in [0, \infty)$, there exists $(x, y) \in X$ such that $\overline{g}([(x, y)]) = t$. Hence, $\overline{g}$ is surjective. Therefore, $\overline{g}$ is bijective.

(iii) $\overline{g}$ is a homeomorphism: We have

$$g = \overline{g} \circ q.$$

Since g is continuous and q is a quotient map, the quotient property of q implies that $\overline{g}$ is continuous.

Also, let U be any open subset of X^*. Since $\overline{g}$ is injective, we have

$$g^{-1}(\overline{g}(U)) = q^{-1}(U).$$

Since q is a quotient map and U is open in X^*, $q^{-1}(U)$ is open in X. Thus, $g^{-1}(\overline{g}(U))$ is open in X. Since g is a quotient map, $\overline{g}(U)$ is open in $[0, \infty)$. Therefore, $\overline{g}$ is an open map.

Hence, $\overline{g}$ is a continuous bijective open map, and consequently it is a homeomorphism. Thus,

$$\boxed{X^* \cong [0, \infty).}$$ □

Problem 7.5

Let $p\colon X \to Y$ be an open map. Show that if A is open in X, then the map $q\colon A \to p(A)$ obtained by restricting p is an open map.

Solution pending

Problem 7.6

Recall that $\mathbb{R}_K$ denotes the real line in the K-topology (see §13). Let Y be the quotient space obtained from $\mathbb{R}_K$ by collapsing the set K to a point; let $p\colon \mathbb{R}_K \to Y$ be the quotient map.

(a) Show that Y satisfies the T_1 axiom, but is not Hausdorff.

(b) Show that $p \times p\colon \mathbb{R}_K \times \mathbb{R}_K \to Y \times Y$ is not a quotient map. [*Hint:* The diagonal is not closed in $Y \times Y$, but its inverse image is closed in $\mathbb{R}_K \times \mathbb{R}_K$.]

Solution pending

Problem Set 8 (Munkres §51, §52)

Problem 8.1

(§51, Ex. 1) Show that if $h, h' \colon X \to Y$ are homotopic and $k, k' \colon Y \to Z$ are homotopic, then $k \circ h$ and $k' \circ h'$ are homotopic.

Solution pending

Problem 8.2

(§51, Ex. 2) Given spaces X and Y, let $[X, Y]$ denote the set of homotopy classes of maps of X into Y.

(a) Let $I = [0, 1]$. Show that for any X, the set $[X, I]$ has a single element.

(b) Show that if Y is path connected, the set $[I, Y]$ has a single element.

Solution pending

Problem 8.3

(§51, Ex. 3) A space X is said to be *contractible* if the identity map $i_X \colon X \to X$ is nulhomotopic.

(a) Show that I and $\mathbb{R}$ are contractible.

(b) Show that a contractible space is path connected.

(c) Show that if Y is contractible, then for any X, the set $[X, Y]$ has a single element.

(d) Show that if X is contractible and Y is path connected, then $[X, Y]$ has a single element.

Solution pending

Problem 8.4

(§52, Ex. 1) A subset A of $\mathbb{R}^n$ is said to be *star convex* if for some point a_0 of A, all the line segments joining a_0 to other points of A lie in A.

(a) Find a star convex set that is not convex.

(b) Show that if A is star convex, A is simply connected.

Solution pending

Problem 8.5

(§52, Ex. 2) Let α be a path in X from x_0 to x_1; let β be a path in X from x_1 to x_2. Show that if $\gamma = \alpha * \beta$, then $\hat{\gamma} = \hat{\beta} \circ \hat{\alpha}$.

Solution pending

Problem 8.6

(§52, Ex. 3) Let x_0 and x_1 be points of the path-connected space X. Show that $\pi_1(X, x_0)$ is abelian if and only if for every pair α and β of paths from x_0 to x_1, we have $\hat{\alpha} = \hat{\beta}$.

Solution pending

Problem 8.7

(§52, Ex. 4) Let $A \subset X$; suppose $r\colon X \to A$ is a continuous map such that $r(a) = a$ for each $a \in A$. (The map r is called a *retraction* of X onto A.) If $a_0 \in A$, show that

$$r_*\colon \pi_1(X, a_0) \longrightarrow \pi_1(A, a_0)$$

is surjective.

Solution pending

Problem 8.8

(§52, Ex. 5) Let A be a subspace of $\mathbb{R}^n$; let $h\colon (A, a_0) \to (Y, y_0)$. Show that if h is extendable to a continuous map of $\mathbb{R}^n$ into Y, then h_* is the trivial homomorphism (the homomorphism that maps everything to the identity element).

Solution pending

Problem 8.9

(§52, Ex. 6) Show that if X is path connected, the homomorphism induced by a continuous map is independent of base point, up to isomorphisms of the groups involved. More precisely, let $h\colon X \to Y$ be continuous, with $h(x_0) = y_0$ and $h(x_1) = y_1$. Let α be a path in X from x_0 to x_1, and let $\beta = h \circ \alpha$. Show that

$$\hat{\beta} \circ (h_{x_0})_* = (h_{x_1})_* \circ \hat{\alpha}.$$

This equation expresses the fact that the following diagram of maps "commutes."

$$\begin{array}{ccc} \pi_1(X, x_0) & \xrightarrow{(h_{x_0})_*} & \pi_1(Y, y_0) \\ \downarrow{\scriptstyle \hat{\alpha}} & & \downarrow{\scriptstyle \hat{\beta}} \\ \pi_1(X, x_1) & \xrightarrow[(h_{x_1})_*]{} & \pi_1(Y, y_1) \end{array}$$

Solution pending

Problem 8.10

(§52, Ex. 7) Let G be a topological group with operation $\cdot$ and identity element x_0. Let $\Omega(G, x_0)$ denote the set of all loops in G based at x_0. If $f, g \in \Omega(G, x_0)$, let us define a loop $f \otimes g$ by the rule

$$(f \otimes g)(s) = f(s) \cdot g(s).$$

(a) Show that this operation makes the set $\Omega(G, x_0)$ into a group.

(b) Show that this operation induces a group operation $\otimes$ on $\pi_1(G, x_0)$.

(c) Show that the two group operations $*$ and $\otimes$ on $\pi_1(G, x_0)$ are the same. [*Hint:* Compute $(f * e_{x_0}) \otimes (e_{x_0} * g)$.]

(d) Show that $\pi_1(G, x_0)$ is abelian.

Solution pending